

SYNTHESIS_v2 — Readable history of human subtelomere research, anchored to the Guarracino & Garrison Nature manuscript

Author: lit-review-v2 (agent-937), 2026-05-06. Supersedes SYNTHESIS.md (lit-review-synthesis, agent-877). **Authors:** paper_prep/synthesis/ABSTRACT_nature.md (primary) and ABSTRACT_BoG.md (talk variant). **Bibliography:** paper_prep/synthesis/REFERENCES_v3.bib (295 entries; merged from REFERENCES.bib, the prior REFERENCES_v2.bib, and the 14 topical topic_NN_.bib files; deduplicated by DOI and (first-author surname, year, normalized-title prefix) with canonical (prefix-free) keys preferred so citations resolve). **Companion files:** CHRONOLOGY_v2.md (timeline table), GAPS_v2.md (open questions and missing-substrate notes), paper_prep/synthesis/CROSSWALK.md (claim ↔ Andrea-report substrate map). **Versioning note:** v1 deliverables (SYNTHESIS.md, CHRONOLOGY.md, GAPS.md, REFERENCES_v2.bib) are retained as historical record. v2 is the canonical reference moving forward; the substantive change from v1 is that topics 05 (acrocentric short arms / rDNA / Robertsonian, refreshed v3-opus pass) and 07 (concerted evolution / NAHR / gene conversion, refreshed v3-opus pass) supply richer per-paper substrate to Parts II and III, and the bibliography now incorporates the final versions of all 14 topical bibs.

Foreword — how to read this document

This document is a single, end-to-end narrative history of the field that the Guarracino & Garrison subtelomere manuscript is entering. It is written for Erik to read in order, before a Board-of-Governors talk or a manuscript-revision session. It is organised into five parts. Parts I–IV each consolidate three to five of the originally-planned topical reviews (01–14) into a chronological field history; transitions between sub-topics are marked in the prose so each part still reads like a unified essay. Part V then steps back to ask how the present claims (C1–C8 of the abstract) sit in this history: which are old discoveries reproduced at population scale, which are new, and which depend on framing decisions that the abstract has chosen.

Citations use BibTeX keys from REFERENCES_v3.bib (e.g. [MeffordTrask2002]); claim labels follow the C1–C8 convention defined in CROSSWALK.md. All 14 topical reviews are committed (topic_01_* cytogenetic foundations, topic_02_* subtelomere structure, topic_03_* pseudohomology concept, topic_04_* sex-chromosome PARs, topic_05_* acrocentric / rDNA / Robertsonian, topic_06_* D4Z4 / DUX4 / FSHD, topic_07_* concerted evolution / NAHR, topic_08_* meiotic bouquet / nuclear-envelope tethering, topic_09_* Hi-C / Pore-C / single-cell / meiotic 3D, topic_10_* pangenome graphs / IMPG, topic_11_* pedigree-based recombination detection, topic_12_* HPRC v1 / v2 population pangenomes, topic_13_* subtelomere population genetics, topic_14_* OR / OR4F gradient) under paper_prep/lit_review/; their bibliographies feed REFERENCES_v3.bib (295 entries) directly. Topics 05 and 07 each received a v3-opus refresh after the v1 synthesis; that refresh is fully integrated below in Parts II and III. The narrative names the canonical primary papers for each part; the topical reviews provide the corresponding per-paper depth and full citation context. GAPS_v2.md records bibliographic and substantive open questions that the synthesis surfaced for future revision passes.

Part I — Cytogenetic and structural foundations (combines topics 01, 02, 04)

From banded karyotypes to fluorescent probes (1916–1990)

The story of human subtelomeres begins, paradoxically, in grasshoppers. In 1916 William Robertson catalogued V-shaped chromosomes in Tettigidae and Acrididae, observed acrocentric–acrocentric fusions and gave the field the word “Robertsonian” that still describes the most common human chromosomal rearrangement involving the five acrocentric short arms (chr13, 14, 15, 21, 22) (Robertson 1916). Acrocentric short arms — and through them, the subtelomeric problem of “what is on the chromosome distal to a heterochromatic block, and how does it move?” — entered human cytogenetics as soon as a banded karyotype existed.

That banded karyotype arrived with quinacrine in 1970. Caspersson, Zech, Johansson and Modest’s quinacrine-banding paper (Caspersson et al. 1970) was the first method that could identify all 24 human chromosomes by their longitudinal pattern; it gave the field G-banding nomenclature and made it possible to talk about chromosome arms (p / q), telomeric and centromeric extremes, and recurrent rearrangements with chromosome-specific resolution. Hsu’s 1979 monograph collected the surrounding cytogenetic foundations (Hsu 1979). Two years after Caspersson, Henderson, Warburton and Atwood used *in situ* hybridisation to localise human ribosomal DNA to the secondary constrictions of all five acrocentric short arms (Henderson, Warburton, and Atwood 1972), proving for the first time that an evolutionarily conserved, highly repetitive, partially homogenised gene family was distributed across multiple non-homologous chromosomes. This is the original “concerted-evolution-at-chromosome-ends” observation in humans, and it underwrites Part II’s pseudohomology concept and Part V’s interpretation of why C5’s acrocentric clade and C8’s “concerted evolution” framing connect at all.

Fluorescence *in situ* hybridisation (FISH) entered the toolset in 1986 with Pinkel, Straume and Gray’s quantitative high-sensitivity protocol (Pinkel, Straume, and Gray 1986), scaled to chromosome painting in Lichter et al.’s 1990 cosmid-FISH demonstration (Lichter et al. 1990) and made systematic in Trask’s 1991 *Trends in Genetics* review (Trask 1991). Together with the discovery of the canonical telomeric (TTAGGG)_n repeat — first characterised in *Tetrahymena* rDNA termini (Blackburn and Gall 1978) and shortly after extended to human chromosome ends — these methods transformed subtelomeres into a measurable rather than theoretical entity. By the early 1990s a researcher could pick a cosmid from a chromosome-end region, tag it with a fluorophore, and ask which other chromosome ends carried homologous sequence in interphase or metaphase nuclei. The Trask lab’s flow-sorted-chromosome FISH programme exploited this: it identified blocks of olfactory-receptor genes duplicated polymorphically among chromosome ends (Trask et al. 1998) and produced cosmid f7501 (L78442), which would later become the field’s single most-revisited subtelomeric block.

In parallel, Brown, MacKinnon, Villasanté and colleagues described “telomere-associated DNA” — *Cell* 1990 — as polymorphic, variably arranged sequence shared across chromosome ends (Brown et al. 1990). This is the founding paper for what the present manuscript calls TAR1 (Telomere-Associated Repeat 1). The next year, Wilkie, Higgs and colleagues used segregation analysis at the tip of chromosome 16p to document a stable length polymorphism of up to 260 kb, which they argued probably arose by occasional exchange among non-homologous subtelomeres (Wilkie et al. 1991). Together these two papers established that chromosome ends were *exchange substrates* rather than fixed reference termini, and they are Part I’s most important pre-1995 claim-anchored finding (it directly motivates C4 — extended interchromosomal homology at nearly all subtelomeres).

The pseudoautosomal region as the prototype (1982–2025)

A decade before the modern molecular era, Burgoyne had already given mammalian cytogenetics the obligate-X/Y-crossover argument: the pseudoautosomal region (PAR1) must recombine in male

meiosis or X and Y will fail to segregate (Burgoyne 1982). The molecular era began the same year as Pinkel's quantitative FISH protocol. In 1986, Goodfellow, Darling, Thomas and Goodfellow showed that *MIC2* (later CD99) is a pseudoautosomal gene with measurable recombination from the sex-determining locus (Goodfellow et al. 1986), and Rouyer, Simmler and colleagues localised the broader phenomenon as a *gradient* of sex linkage (Rouyer et al. 1986): markers very near the chromosome tip behaved as if autosomal (high recombination; low LD with sex), while markers proximal to the PAR boundary behaved as sex-linked. Pritchard et al. used long-range restriction mapping to place *MIC2* close to Y-specific sequence (Pritchard, Goodfellow, and Goodfellow 1987), and Ellis, Goodfellow and colleagues then identified an Alu-repeat insertion at the proximal PAR1 boundary on the Y chromosome (Ellis et al. 1989; 1990) — the first molecular description of an abrupt boundary between recombining and non-recombining sex-chromosome territory. PAR1 thus became the first explicitly characterised “pseudohomologous” region in the human genome, with an obligate functional role tying physical proximity at the chromosome tip to recombination.

The pseudoautosomal frame expanded in the 1990s. Freije, Helms, Watson and Donis-Keller used microsatellite inheritance in CEPH pedigrees to discover a *second* pseudoautosomal region near the Xq/Yq tips (Freije et al. 1992); Graves, Wakefield and Toder synthesized comparative-mammal evidence that PAR1 and PAR2 have different evolutionary origins (PAR1 as relic added autosomal material, PAR2 as a more recent compound event) (Graves, Wakefield, and Toder 1998); Lahn and Page placed both PARs alongside the X-conserved (XCR), X-added (XAR) and X-transposed (XTR) regions in their evolutionary-strata model (Lahn and Page 1999). Filatov then demonstrated a gradient of silent substitution rate across the human PAR consistent with the sex-averaged recombination gradient (Filatov 2004). The sequence era refined both ends: Ciccodicola et al. produced the first full sequence of the Xq/Yq pseudoautosomal interval and showed it has internally distinct gene-regulation and evolutionary subregions (Ciccodicola et al. 2000); Charchar et al. mapped PAR2 as arising through multiple species-specific events rather than a simple single duplication (Charchar et al. 2003); Skaletsky et al.'s Y-chromosome reference assembly placed PAR1, PAR2, X-degenerate, X-transposed and ampliconic territory in one mosaic (Skaletsky et al. 2003); and Ross et al.'s X-chromosome reference completed the picture by naming PAR1 / PAR2 / XAR / XCR / XTR in one framework (Ross et al. 2005). Mangs and Morris consolidated this into a current-genomics review (Mangs and Morris 2007); Flaquer, Rappold, Wienker and Fischer wrote the genetic-epidemiology PAR primer (Flaquer et al. 2008) and the sex-specific genetic-map paper that quantified the contrast between PAR1 (extreme male recombination) and PAR2 (near-genome-average) (Flaquer, Fischer, and Wienker 2009). PAR1 is approximately 2.7 Mb and supports the obligate X–Y crossover; PAR2 is roughly 320–400 kb, has a different origin, and undergoes much less recombination.

From 2014 onward the PAR literature shifted from “where are the PARs?” to “how stable are PAR boundaries, how much sequence is required for function, and what does direct recombination mapping look like at the molecular level?” Hinch, Altomose, Noor, Donnelly and Myers produced the canonical fine-scale recombination map of PAR1 from pedigrees and LD, with intense male recombination and resolvable hotspot structure (Hinch et al. 2014). Mensah et al. then identified recurrent human PAR1 length polymorphism caused by X-to-Y translocation — the first ePAR description (Mensah et al. 2014); Cotter, Brotman and Wilson Sayres showed that population-genetic diversity does not produce a clean cliff at the annotated PAR1 boundary, supporting boundary plasticity (Cotter, Brotman, and Wilson Sayres 2016); Poriswanish et al. characterised recombination hotspots within an extended human pseudoautosomal domain by sperm-based crossover analysis (Poriswanish et al. 2018); Acquaviva et al. revealed in mouse that ANKRD31 and MEI4-related axis-elongation factors force DSB formation into the pseudoautosomal segment so that even a small interval can carry the obligate exchange (Acquaviva et al. 2020); Fukami et al. provided pedigree

evidence that human spermatogenesis tolerates massive size reduction of PAR1 (Fukami et al. 2020); Monteiro, Arenas, Prata and Amorim built the population-genetic dynamics model for PARs (Monteiro et al. 2021); Bergman and Schierup placed PAR1 in great-ape population-genomic context with recombination-associated mutation and GC-biased gene conversion (Bergman and Schierup 2022). The T2T era arrived with Miga et al.'s telomere-to-telomere X chromosome (Miga et al. 2020), Rhie et al.'s complete Y (Rhie et al. 2023) and Hallast et al.'s 43 assembled human Y chromosomes documenting extensive PAR1 length polymorphism in the population (Hallast et al. 2023); and Bellott, Hughes, Skaletsky, Owen and Page re-examined genetic and sequence evidence to narrow the canonical PAR1 boundary against a competing 2023 assembly-based shift claim (Bellott et al. 2024). Most recently Poriswanish et al. analysed approximately 218,300 46,XY males in UK Biobank, estimating ePAR1 incidence, recurrent origins across multiple Y haplogroups, reciprocal X deletions and phenotypic associations (Poriswanish et al. 2025) — establishing that PAR1 is not a fixed textbook interval but a recombination *system* with recurrent structural variants creating new pseudoautosomal sequence in living populations.

The result is a clean prototype: PAR1 and PAR2 are bona fide pseudohomologous regions; their recombination behaviour is functionally required for sex-chromosome segregation or for evolutionary turnover of male-specific gene content; their boundaries are biologically plastic at population scale; and their existence proves that interchromosomal sequence sharing at chromosome ends is biologically tolerated — even *required* — at the human germline level. This is the empirical anchor for C4's "comparable in scale to the canonical PAR2" wording: when the present manuscript reports tens-to-hundreds of kilobases of homology at nearly all subtelomeres comparable to PAR2, the reader is being asked to view PAR2 not as the only such region but as one named instance of a much broader phenomenon. It is also the empirical anchor for C5's positive controls: the manuscript's NJ-tree clades C14 (Xq/Yq → PAR2) and C15 (Xp/Yp → PAR1) are recovering the two best-known sex-chromosome subtelomeric clades. Per the topic_04 review's interpretation, PAR1 supplies the *strong* mechanistic anchor (genetically mapped, sperm-typed obligate male crossover) and PAR2 the *weaker* anchor (true Xq/Yq homology but with crossover rate near genome average); together they calibrate the manuscript's broader autosomal-PHR claims by giving two named instances where recombination is *already* part of the definition.

Subtelomeric structural biology: TAR1, internal telomeric sequences and the two-domain model (1996–2008)

Brown's TAR1 (Brown et al. 1990) and Wilkie's chr16p polymorphism (Wilkie et al. 1991) launched a structural-biology programme that asked what the subtelomeric repeat landscape actually looks like at the sequence level. The 1996 sequencing of the chr4q35 D4Z4 macrosatellite (Deutekom et al. 1996) and the 1992 mapping of 4q-tip rearrangements to FSHD (Wijmenga et al. 1992) gave the field a disease-linked exemplar of subtelomeric exchange — the chr4q ↔ chr10q D4Z4 translocation observed in roughly 20% of typed individuals (Deutekom et al. 1996). Flint, Bates and colleagues then proposed the two-domain model in 1997 (Flint et al. 1997): a distal subtelomeric domain that shares sequence with many chromosome ends, and a more proximal domain shared with fewer ends, separated by degenerate internal (TTAGGG)_n tracts. This model was inferred from comparison of human and yeast subtelomeres and was characterised at chr4p, chr16p and chr22q only; it predicts both that interchromosomal sharing should *decrease* with distance from the telomere and that internal telomeric sequence islands should sit at the boundary. It is the single most-tested prediction of the Flint era and is precisely what the present manuscript's "two-domain Flint/Mefford model" result re-tests at pangenome scale (39/48 arms gradient; 39/41 piecewise-breakpoint; 11/19 ITS within 25 kb of the breakpoint — a confirmation orders of magnitude broader than the original three arms).

Internal (TTAGGG)_n repeats themselves became a research object. Ruiz-Herrera, Nergadze and colleagues reviewed the mechanisms by which telomeric tracts end up far from chromosome ends — interstitial telomeric sequences (ITS) — and the conserved evolutionary signal they carry (Ruiz-Herrera et al. 2008). By 2008 the field accepted that subtelomeres contain a stable architecture of (i) a very distal canonical (TTAGGG)_n telomere; (ii) TAR1 (Brown 1990) immediately proximal; (iii) a duplicated patchwork of subtelomere-specific repeats; (iv) often, internal (TTAGGG)_n islands deeper in. In the present manuscript, chapters 02 and 04 of Andrea's report quantify each of these layers across 466 haplotypes (TAR1 prevalence 94.6%; absent from PAR1 at 0.5%; 18,352 internal (TTAGGG)_n islands across 8,321 sequences), and these observations are direct population-scale validation of the Flint and Brown structural model.

Riethman, Ambrosini and the assembly era (2001–2014)

Riethman, Xiang and colleagues integrated telomere clones with the draft human genome in 2001 (Riethman et al. 2001) and established the systematic subtelomeric assembly programme that would dominate the subsequent decade (Riethman et al. 2004). The conceptual upshot was that ~80% of the most distal 100 kb of human chromosomes consists of shared duplicated blocks; subtelomeres are not single-copy regions punctuated by telomeres but rather *patchworks* of duplications. Ambrosini, Paul, Hu and Riethman then catalogued these patchworks in 2007 as a duplicon-block organisation (Ambrosini et al. 2007): 11 subtelomere-specific block families, 6 subterminal block families, 6 one-copy (TTAGGG)_n-adjacent regions, and a bimodal sequence-identity distribution with peaks at ~91% and ~98% (the two peaks reflecting older and more recent exchange events respectively). Riethman's 2008 review then placed copy-number variation at subtelomeres in the broader CNV literature (Riethman 2008). Stong, Deng and colleagues' 2014 paper extended the assembly programme into structural and chromatin annotation (Stong et al. 2014), establishing CTCF and cohesin organisation across haplotype-resolved subtelomeres.

The assembly era is conceptually important for two reasons. First, it converted the FISH-era observation that subtelomeres exchange into a *catalogue* of which sequence blocks exchange between which chromosome ends — vocabulary the present manuscript inherits when it names communities (C1 D4Z4-bearing 4q–10q, C2 chr10p–chr18p, C3 f7501-cluster, C5 RPL23A/WASH, etc.). Second, it exposed the limits of single-haplotype references: even with a full-length BAC tile across a chromosome-end region, individual subtelomeres show major copy-number, allele-organisation and gene-content variation that a reference cannot represent. This exposure motivates C2 (the implicit pangenome graph as a reference-free response) and C3 (466 near-complete haplotype assemblies as the substrate that finally makes the assembly problem tractable across populations).

Long-molecule and T2T eras (2018–2022)

Optical mapping and long-read sequencing reopened subtelomere assembly from the technology side. Shao, Zhou and colleagues (Shao et al. 2018) used BioNano and nanopore data to detect ongoing chromosome-end extension events — heterozygous extensions that could be modelled as chimeric chromosome ends. Young, Abid and colleagues' whole-genome optical mapping of 154 individuals showed widespread large-scale subtelomeric structural variation undersampled in standard references (Young et al. 2020). Grigorev, Foox and colleagues used long-read telomere sequencing to characterise telomeric repeat-motif heterogeneity across individuals (Grigorev et al. 2021). Adam, Ranjan and Riethman's NPGREAT pipeline integrated linked reads with ultralong nanopore reads to assemble CHM13 subtelomeres across representative segmental-duplication structures (Adam, Ranjan, and Riethman 2022). Logsdon et al.'s complete chr8 (Logsdon et al. 2021) and Nurk et al.'s T2T-CHM13 assembly (Nurk et al. 2022) then closed many formerly inaccessible terminal regions; Vollger et al. catalogued the segmental-duplication landscape at T2T scale (Vollger

et al. 2023). The HPRC v1 draft pangenome demonstrated that haplotype-resolved diverse assemblies could be organised into a population reference (Liao et al. 2023).

This sequence of advances is the technical bridge to C1–C3 of the present manuscript. The 466 HPRC v2 near-complete haplotype assemblies are the operational endpoint of three decades of subtelomeric structural biology: each arm is now a sequenced haplotype rather than an inferred FISH signal, each haplotype is part of a population-scale collection rather than a single reference, and each subtelomere can be aligned all-to-all without imposing chromosomal partitioning. Part I of this synthesis ends here, at the moment in 2022 when reading every subtelomere in 233 individuals became practical. Parts II–IV pick up the conceptual, mechanistic and methodological histories that turn that data into the manuscript’s claims.

Part II – The pseudohomology concept and its variants (combines topics 03, 05, 06, 13, 14)

The pseudohomology concept proper (1990–present)

The central conceptual unit of this manuscript is the **pseudohomologous region (PHR)** – telomere-adjacent sequence that is paralogously shared across non-homologous chromosome ends, in contrast to the formal *pseudoautosomal* region (PAR) which is functionally homologous between X and Y. The PHR concept emerged in three layers, each documented in detail in the topic_03 review [paper_prep/lit_review/topic_03_pseudohomologous_regions_concept.md] and summarised here.

The first layer is *architectural*: Flint et al.’s two-domain model (Flint et al. 1997) proposed that chromosome ends share a distal multi-end domain and a proximal few-end domain separated by internal (TTAGGG)_n tracts. The second layer is *evolutionary*: Trask 1998 (Trask et al. 1998), Mefford & Trask 2002 (Mefford and Trask 2002), Der-Sarkissian et al. 2002 (Der-Sarkissian et al. 2002), Linardopoulou et al. 2005 (Linardopoulou et al. 2005) and Ambrosini et al. 2007 (Ambrosini et al. 2007) turned the architecture into a dynamic patchwork of duplicated blocks moved among ends at high rate. Mefford & Trask in particular framed subtelomeres as transitions between chromosome-specific DNA and terminal telomere repeats with recurrent exchange and copy-number variation, and explicitly hedged that block frequencies might carry population-history signal even though exchange history could make them unreliable phylogenetic markers – a hedge that the present manuscript’s chapter 04 + chapter 12 results revisit by recovering the out-of-Africa topology from cross-arm-affinity Fst (chapter 12 novel contribution #19; supports C6). The third layer is *operational*: Riethman 2004 (Riethman et al. 2004), Stong 2014 (Stong et al. 2014), Adam 2022 (Adam, Ranjan, and Riethman 2022), Nurk 2022 (Nurk et al. 2022) and Liao 2023 (Liao et al. 2023) made it possible to *measure* PHRs at sequence resolution across many haplotypes – which is the technical premise of C2 (implicit pangenome graph) and C3 (466 haplotypes).

The PHR concept is also bounded by alternatives. Bailey, Gu and Eichler’s “recent segmental duplications” framework (Bailey et al. 2002), updated by Eichler 2001 (Eichler 2001), Sharp 2006 (Sharp et al. 2006), Sudmant 2015 (Sudmant et al. 2015) and Vollger 2023 (Vollger et al. 2023), placed subtelomeric exchange within a genome-wide segmental-duplication story rather than a chromosome-end-specific phenomenon. Stankiewicz & Lupski’s NAHR / non-allelic homologous recombination framework (Stankiewicz and Lupski 2002; 2010) provided the mechanistic vocabulary of *how* large duplicated repeats mediate ectopic recombination through misalignment at homologous templates. Lovett’s misalignment-mediated mutation review (Lovett 2004) supplied the molecular detail for slipped-strand and template-switching events that produce sub-block variation. And Rudd, Friedman et al. 2007 (Rudd et al. 2007) showed that even sister-chromatid exchange rates are

elevated at chromosome ends, which is a cell-biological reason chromosome ends accumulate exchange events broadly, not merely through one mechanism.

The current manuscript inherits and refines this concept. The key operational definition is: a PHR is the telomere-adjacent interval of a haplotype that shares high-identity sequence with other chromosome ends in the implicit pangenome graph; communities are inferred from the resulting similarity graph rather than imposed from cytogenetic labels [paper_prep/lit_review/topic_03_pseudohomologous_regions_concept.md, lines 31–35]. The novelty of C4–C5 is strongest where all three historical layers converge: the old two-domain and patchwork models are recovered, known clades (PAR1, PAR2, acrocentric, 10p–18p, f7501-cluster, 4q–10q D4Z4) are reproduced, novel clades appear, and the population-scale structure becomes measurable for the first time.

The acrocentric short arms and rDNA (topic 05 — refreshed v3-opus pass)

The five acrocentric short arms (chr13, 14, 15, 21, 22) are the single best-studied “concerted-evolution at chromosome ends” example, and the v3-opus refresh of topic_05 substantially deepens the substrate available to the manuscript. Henderson, Warburton and Atwood’s 1972 *in situ* hybridisation (Henderson, Warburton, and Atwood 1972) showed rDNA arrays distributed across all five short arms; Schmickel 1973 (Schmickel 1973) quantified human rDNA copy number at roughly 200 per haploid genome, the first dosage estimate for the five-site cistrons. Krystal et al. 1981 (Krystal et al. 1981) used somatic-cell hybrids to show that nonhomologous human acrocentric chromosomes carry the same intergenic-spacer length variants — direct molecular evidence that sequence information moves between heterologous NORs and a foundational test of concerted evolution; Worton et al. 1988 (Worton et al. 1988) mapped the orientation of the rDNA tandem arrays; Therman, Susman and Denniston 1989 (Therman, Susman, and Denniston 1989) catalogued Robertsonian translocations and noted the dominance of chr13–14 and chr14–21 fusions in the empirical distribution. The molecular mapping era then narrowed the breakpoints: Page, Shaffer and Shaffer 1996 (Page and Shaffer 1996) localised rob(13;14) and rob(14;21) breakpoints to the satellite-III / rDNA boundary; Bandyopadhyay et al. 2001 (Bandyopadhyay et al. 2001) identified seven new satellite III subfamilies across the five p-arms; Bandyopadhyay et al. 2002 (Bandyopadhyay et al. 2002) established that common Robertsonian translocations are predominantly dicentric, occur preferentially in female meiosis, and may be driven by nucleolar association. Stults et al. 2008 (Stults et al. 2008) then used optical pulsed-field gel electrophoresis to measure rDNA cluster lengths across multigenerational pedigrees, reporting a meiotic rearrangement frequency exceeding 10% per cluster per meiosis — among the highest documented rates of structural change in the human genome.

Floutsakou and colleagues’ 2013 paper (Floutsakou et al. 2013) then sequenced the *distal junction* (DJ) and *proximal junction* (PJ) regions flanking the rDNA on chromosome 21, found that PJ and DJ are 95% and 99% identical across the five acrocentrics, and identified the distal-junction long noncoding transcript that orchestrates the surrounding chromatin. Stimpson et al. 2014 (Stimpson et al. 2014) linked nucleolar organisation, rDNA stability and acrocentric integrity to telomere function. Jarmuz-Szymczak et al. 2014 (Jarmuz-Szymczak et al. 2014) further refined Robertsonian breakpoints using FISH on *de novo* and familial cases. McStay’s 2016 review consolidated these findings into the “nucleolar organiser regions are genomic dark matter” frame (McStay 2016); van Sluis et al. 2019 extended the architecture and confirmed near-identity of acrocentric short arms in human and chimpanzee (Sluis et al. 2019). Most recently, Guarracino et al. 2023 (Guarracino et al. 2023) used pangenome-graph methods on HPRC v1 data to demonstrate ongoing recombination between heterologous acrocentric short arms — the canonical predecessor of the present manuscript and the paper that establishes “subtelomeric exchange across non-homologous chromosomes is detectable in pangenome graphs.” de Lima, Guarracino et al. 2025 (Lima, Guarracino, and others 2025) completed the assemblies of three Robertsonian chromosomes (rob(13q14q), rob(14q21q)) and

identified the SST1 macrosatellite array — inverted on chromosome 14 — as the recombining homology that produces the dominant fusion product. Cechova, Hartley et al. 2025 (Cechova et al. 2025) extended the analysis across complete great-ape acrocentric assemblies, showing recurrent segmental duplication, FRG1 amplification and centromere repositioning across 25 Myr of ape evolution. Hartley et al. 2026 (Hartley et al. 2026) developed a short-read genotyping method for ROB carriers and applied it to UK Biobank (~490 k samples) plus a healthy newborn cohort, recovering the historical 0.11–0.12% prevalence and uncovering hidden distal-junction copy-number variation in HPRC genomes.

For the present manuscript these papers are the substrate of *C5's acrocentric clade* (chapter 01 community C7) and of the “concerted evolution” framing in C8: the rDNA-adjacent acrocentric homogenisation is the most extreme case of within-community paralog distance < allelic distance (chapter 04: C7 reverses with paralog closer $p=2.0e-7$, chapter 12 novel contribution #18). Concerted evolution is *the* canonical mode of evolution for the rDNA gene family (Dover 1982; Smith 1976; Liao 1999; Hurles 2004); the present manuscript extends that mode beyond rDNA into the surrounding subtelomeric blocks. Critically, de Lima 2025's identification of the SST1 macrosatellite as the recombining homology supplies the *molecular* mechanism for the dominant Robertsonian products — direct evidence that ongoing exchange among the five short arms is mediated by specific homologous templates, not by generic NOR-clustering alone.

D4Z4, FSHD biology and the chr4q–chr10q axis (topic 06)

The D4Z4 macrosatellite repeat at the 4q35 subtelomere is the second canonical example. Wijmenga et al. 1992 mapped 4q-tip rearrangements to facioscapulohumeral muscular dystrophy (Wijmenga et al. 1992); van Deutekom et al. 1996 (Deutekom et al. 1996) showed Southern-blot evidence of subtelomeric exchange of 3.3 kb tandem D4Z4 units between chr4q35 and chr10q26 in roughly 20% of typed individuals; Lemmers et al. 2002 (Lemmers et al. 2002) established that FSHD is associated with one of two 4q subtelomere variants (4qA / 4qB); Masny et al. 2004 (Masny et al. 2004) localised 4q35.2 to the nuclear periphery via lamin A/C; Kowaljow et al. 2007 (Kowaljow et al. 2007) showed that DUX4 within D4Z4 encodes a pro-apoptotic protein; Ottaviani et al. 2008 (Ottaviani et al. 2009a) identified a perinuclear positioning element in subtelomeres that requires A-type lamins and CTCF; Ottaviani et al. 2009 (Ottaviani et al. 2009b) then established that D4Z4 itself acts as a CTCF / A-type lamin-dependent insulator; Snider et al. 2010 (Snider et al. 2010) showed FSHD reflects incomplete suppression of a retrotransposed gene; Lemmers et al. 2010 (Lemmers et al. 2010) gave the unifying genetic model — permissive 4qA haplotype produces stable DUX4 mRNA via a 4q-specific polyadenylation signal; Cebianca et al. 2012 (Cebianca et al. 2012) connected D4Z4 copy-number to a long ncRNA-mediated polycomb/trithorax epigenetic switch.

This dense literature is the molecular underpinning of *C5's chr4q–chr10q DUX4 community* (chapter 01 community C1, named for the D4Z4 sharing) and *C7's nuclear-envelope hypothesis* (the abstract's “facilitated by the physical proximity of subtelomeres at the nuclear envelope” wording derives directly from the Masny–Ottaviani D4Z4-CTCF-lamin axis). The present manuscript's chapter 03 confirms 28 DUX4L genes in C1, chapter 07 cites the Masny–Ottaviani mechanism in the integrated synthesis, and chapter 14 detects a chr10q ← chr4q gene-conversion event at score 0.957 in PAN028 maternal — a *directly observed* subtelomeric exchange event in the WashU pedigree confirming the FSHD-context translocation as ongoing rather than only inferable from population frequency.

Subtelomere population genetics and out-of-Africa structure (topic 13)

The pseudohomology concept's third population-genetics dimension is whether subtelomeric block frequencies carry phylogenetic signal. Mefford & Trask 2002 raised this directly, hedging that “measuring the frequency of specific subtelomeric blocks on particular chromosomes might inform

phylogenetic studies of modern humans” but cautioning that “such frequencies could be an unreliable indicator of the relationships of human populations” because they reflect exchange history as well as population history (Mefford and Trask 2002). Der-Sarkissian et al. 2002 (Der-Sarkissian et al. 2002) supplied early evidence that proterminal markers vary across populations.

The decade of population-genomics that followed (1000 Genomes Project Consortium 2015 (1000 Genomes Project Consortium 2015); Sudmant et al. 2015 SV map (Sudmant et al. 2015); Mallick et al. 2016 SGDP (Mallick et al. 2016); Bergström et al. 2020 HGDP (Bergström et al. 2020)) made it possible to test Mefford’s hedge: do subtelomeric block frequencies, treated as alleles, recover the canonical out-of-Africa topology? Andrea’s chapter 04 (Hudson Fst) and chapter 12 (novel contribution #19) report that they do. From cross-arm affinity Fst at 9 arms × 5 superpopulations, AFR is the deepest split (distance 0.73–0.81), AMR–EUR are closest (distance 0.22), the topology mirrors out-of-Africa, and AFR/non-AFR Fst is 0.10–0.15 while non-AFR pairs are 0.00–0.02. This is the direct quantitative answer to Mefford & Trask’s 2002 hedge: subtelomeric blocks *do* carry phylogenetic signal at population scale, and the signal is large enough to recover deep splits. It supports C6 (PCA + community detection on similarity matrix → subtelomere clustering across human populations) and provides one of the strongest novel contributions of the manuscript — chapter 12 novel contribution #19.

Olfactory-receptor pseudogenisation and the f7501 / OR4F gradient (topic 14)

Olfactory receptors are the largest mammalian gene family and the canonical example of a duplicating, pseudogenising gene family at chromosome ends. Glusman et al. 2001 catalogued the complete human olfactory subgenome (Glusman et al. 2001); Niimura & Nei 2003 documented evolutionary dynamics (Niimura and Nei 2003); Olender et al. 2008 published a structured update (Olender, Lancet, and Nebert 2008); Hasin-Brumshtein, Lancet and Olender 2009 connected genomic variation to phenotypic diversity (Hasin-Brumshtein, Lancet, and Olender 2009); Niimura, Matsui and Touhara 2014 placed the human OR repertoire in a 13-mammal comparative frame (Niimura, Matsui, and Touhara 2014). Trask 1998 (Trask et al. 1998) is the founding empirical observation that OR cluster blocks are duplicated polymorphically near chromosome ends in humans; cosmid f7501 (L78442) was the field’s reference probe for one such block.

The Mefford & Trask 2002 review (Mefford and Trask 2002) documented f7501 distribution across ~52 individuals at five chromosome ends (chr3q, chr15q, chr19p, chr16q, chr7p) and found chr16q enriched in African populations. Andrea’s chapter 01 §“f7501 (L78442) direct alignment validation” reproduces this *quantitatively* at population scale: chr3q (91.8%), chr19p (90.5%), chr15q (85.6%) confirmed *fixed* across the 466-haplotype dataset; chr16q AFR-enriched at $p=6.6e-27$ (70% AFR carriers vs <10% non-AFR); novel SAS-enriched chr2q ($p=6.8e-11$), AMR-enriched chr6p ($p=7.0e-04$), and three additional AFR-enriched arms (chr8p, chr16p, chr9q). Chapter 03 then reports that 62.1% of 5,023 OR4F annotations are pseudogenes, with pseudogenisation rate varying from 11.1% (chr7p) to 99.8% (chr15q) — the “OR4F gradient” finding in the C5 community structure (chapter 12 novel contribution implicit in the OR4F→C8 / DUX4L→C1 / IQSEC3→C5 / IL9RP1→C3 mapping table from chapter 03).

For the present manuscript, the OR-family literature is critical context for two reasons. First, OR pseudogenisation gradients across communities give the C5 cladistic structure a *mechanistic* interpretation beyond pure sequence affinity: communities differ in the rate at which OR genes lose function as they move across chromosome ends, suggesting different selective regimes at different chromosome ends. Second, the f7501 reproduction in chapter 01 is the single cleanest “old discovery confirmed at scale” result in the manuscript and should be foregrounded in any BoG Q&A on the validity of community-detection methods.

Summary — pseudohomology, four ways

By the end of Part II the field knows four kinds of pseudohomology. (i) PAR-style obligate homology — chrX/Y PAR1 and PAR2, where exchange is functionally required and the gradient of sex linkage (Rouyer et al. 1986; Goodfellow et al. 1986; Freije et al. 1992; Filatov 2004; Charchar et al. 2003; Hinch et al. 2014; Bellott et al. 2024) is the canonical signature, with recurrent population-scale length polymorphism establishing that PAR1 is a recombination *system* rather than a fixed interval (Mensah et al. 2014; Poriswanish et al. 2018; 2025); this is Part I material but is the cleanest prototype. (ii) Acrocentric rDNA-flanking concerted evolution — the chr13/14/15/21/22 short arms, where homogenisation extends across the rDNA and into the surrounding subtelomere; canonical predecessor papers (Henderson, Warburton, and Atwood 1972; Floutsakou et al. 2013; McStay 2016; Sluis et al. 2019; Guarracino et al. 2023; Lima, Guarracino, and others 2025); the present manuscript's C7 community. (iii) Block-shared dynamic patchworks — the FISH-era / Linardopoulou / Ambrosini concept of recurrent ectopic exchange among subtelomeric blocks at non-acrocentric ends (Mefford and Trask 2002; Linardopoulou et al. 2005; Ambrosini et al. 2007); this is what most of the present manuscript's communities (C1, C2, C3, C5, C6) represent. (iv) Disease-associated subtelomeric duplicons — D4Z4 / FSHD at chr4q-chr10q (Wijmenga et al. 1992; Deutekom et al. 1996; Lemmers et al. 2010) — a single named case where pseudohomology has clinical weight. The C5 cladistic claim of the abstract names instances of all four kinds; the C8 “concerted evolution” framing is the umbrella across all four. Part III now asks the mechanistic question: what is the cellular and molecular machinery that creates and maintains this pseudohomology?

Part III — Mechanism: meiotic 3D and recombination (combines topics 07, 08, 09)

Concerted evolution, gene conversion and NAHR (topic 07 — refreshed v3-opus pass)

The concerted-evolution concept enters the field with Smith 1976 (Smith 1976), who showed that unequal crossover among tandem repeats generates a homogenisation dynamic, and Brown 1972 (Brown, Wensink, and Jordan 1972) who had already demonstrated the empirical pattern in *Xenopus* rDNA. Dover 1982 (Dover 1982; 1986) gave the field the *molecular drive* framework — a cohesive mode of species evolution in which intragenomic processes (gene conversion, unequal crossover, transposition) homogenise repetitive sequence families across the genome. Liao 1999 (Liao 1999) consolidated the concerted-evolution literature and articulated unequal crossover and gene conversion as the two principal molecular engines; Hurles 2004 (Hurles 2004) supplied the CMT1A paralogous-repeat gene-conversion exemplar. Nei and Rooney 2005 (Nei and Rooney 2005) introduced the *birth-and-death* alternative for some multigene families — an important caveat for OR4F (where the family does turn over rather than being homogenised) and a reason the abstract uses “concerted evolution” only in the loose sense.

The NAHR / genomic-disorders frame entered with Lupski 1998 (Lupski 1998) and was canonised by Stankiewicz & Lupski 2002 / 2010 (Stankiewicz and Lupski 2002; 2010), with the broader genomic-disorders synthesis built up across Lupski & Stankiewicz 2005 (Lupski and Stankiewicz 2005) and Lupski 2009 (Lupski 2009). Bailey & Eichler 2006 (Bailey and Eichler 2006) argued that primate segmental duplications are evolutionarily young, structurally biased toward pericentromeric and subtelomeric locations, and recombinationally active — directly relevant to subtelomeres as a high-NAHR class. Chen, Cooper and colleagues 2007 (Chen et al. 2007) reviewed gene conversion mechanisms and human disease; Hastings et al. 2009 (Hastings et al. 2009) integrated NAHR, NHEJ and FoSTeS/MMBIR into a single mechanism-of-CNV-change synthesis. Lovett 2004 (Lovett 2004) supplied the misalignment / template-switching mechanism at the nucleotide level. Bailey 2002 (Bailey et al. 2002) and Sharp 2006 (Sharp et al. 2006) established that recent segmental duplications

are the genome-wide arena for these mechanisms, and Sudmant 2015 (Sudmant et al. 2015) / Vollger 2023 (Vollger et al. 2023) placed segmental duplication in the population-scale and T2T-scale context.

The third mechanism — break-induced replication (BIR) — joined the inventory through Llorente, Smith and Symington 2008 (Llorente, Smith, and Symington 2008) and Anand, Lovett and Haber 2013 (Anand, Lovett, and Haber 2013), which showed that long-tract synthesis from a single homologous template can copy hundreds of kilobases without crossover. BIR is the operationally relevant mechanism for the kilobase-scale gene-conversion patches the present manuscript reports: the WashU pedigree's 538 patches range from a few kb up to hundreds of kb, and 133 are gene-conversion-like (no crossover) at perfect 1.000/1.000 score — a size distribution and exchange morphology more compatible with BIR than with classical short-tract gene conversion.

The intra-array biology of rDNA — the strict-sense concerted-evolution case — was quantified by Eickbush & Eickbush 2007 (Eickbush and Eickbush 2007) and Ganley & Kobayashi 2007 (Ganley and Kobayashi 2007): total repeat homogenisation in real time, mechanism mapped onto recombination at the ribosomal locus. Galtier et al. 2001 (Galtier et al. 2001) and Duret & Galtier 2009 (Duret and Galtier 2009) established that gene conversion is GC-biased, with population-genetic and isochore consequences; gBGC is a candidate explanation for why subtelomeric communities tend to retain GC-rich block structure. Jeffreys & May 2004 (Jeffreys and May 2004) measured human gene-conversion tract length at ~50–500 bp from sperm crossover hotspots; Williams et al. 2015 (Williams et al. 2015) and Halldorsson et al. 2019 (Halldórsson et al. 2019) produced direct pedigree-resolved gene-conversion calls at sequence resolution, providing the closest external comparator for the WashU pedigree analysis.

For the present manuscript, two threads matter most. First, NAHR is the canonical mechanism for *crossover-like* PHR exchange events (chapter 14 reports 16 such events in the WashU pedigree), and gene conversion (with BIR as a long-tract variant) is the mechanism for the *gene-conversion-like* events (chapter 14: 133 events, 96 of which sit at perfect 1.000/1.000 score). Second, “concerted evolution” in the abstract title refers — per the lead author's clarification — to the *loose* sense (ongoing recombination exchange producing sequence-similarity homogenisation across non-homologous arms), not the strict molecular-evolution sense (a coherent, often co-evolutionary process across a gene family). Nei & Rooney 2005 (Nei and Rooney 2005) is the explicit reason for this loose-sense framing: not all subtelomeric paralog families behave like rDNA, and the loose sense covers both homogenising (rDNA) and turnover-prone (OR4F) systems. The pedigree result (Part IV) is what upgrades that loose sense from inference to direct observation.

The meiotic bouquet and nuclear-envelope tethering (topic 08)

The meiotic bouquet is the central cell-biological observation that ties subtelomeric sequence homology to chromosome positioning. Scherthan, Weich and colleagues 1996 (Scherthan et al. 1996) showed that centromere and telomere movements during early meiotic prophase of mouse and man are associated with the onset of chromosome pairing. Zickler & Kleckner's 1998 review (Zickler and Kleckner 1998) consolidated the leptotene-zygotene transition as the key window; their 2015 *Cold Spring Harbor Perspectives* article (Zickler and Kleckner 2015) is the canonical modern review of recombination, pairing and synapsis with the LINC complex / SUN-KASH machinery framing.

The molecular machinery is now well-characterised. Sosa, Rothballer and colleagues 2012 (Sosa et al. 2012) solved the structural basis of LINC complex formation through KASH-peptide / SUN-trimer binding; Penkner et al. 2009 (Penkner et al. 2009) identified the SUN-1 modification cycle that drives meiotic chromosome homology search through the nuclear envelope; Hiraoka & Dernburg 2009 (Hiraoka and Dernburg 2009) reviewed the SUN-protein meiotic chromosome dynamics; Chikashige et al. 2006 (Chikashige et al. 2006) identified the fission-yeast Bqt1/Bqt2 telomere tethers; Lee et

al. 2015 (Lee et al. 2015) worked out rapid mouse telomere prophase movements; Boateng, Bellani and Camerini-Otero 2013 (Boateng et al. 2013) documented homologous pairing preceding SPO11-mediated double-strand breaks in mice. Burgoyne 1982 (Burgoyne 1982) is the original X/Y obligate-crossover argument that anchors the C7 hypothesis. The mouse-PAR mechanism extends this further: Acquaviva, Boekhout, Karasu et al. 2020 (Acquaviva et al. 2020) showed that ANKRD31, MEI4 and axis-elongation factors actively *force* DSB formation into the small pseudoautosomal interval to ensure the obligate X-Y crossover, and Chu, Froberg, Kesner et al. 2017 (Chu et al. 2017) identified PAR-TERRA RNA as a pairing factor for sex-chromosome homologous regions in mouse ESCs. Together these papers establish that chromosome-end pairing is *actively maintained* by sequence-aware molecular machinery, not merely a passive consequence of telomere clustering.

Two takeaways matter for C7 and C8. First, the meiotic bouquet places telomere-adjacent regions at the nuclear envelope, in physical proximity to one another, during a window (leptotene→zygotene) in which homology search and double-strand-break repair are most active. This is the biophysical basis for the C7 “facilitated by the physical proximity of subtelomeres at the nuclear envelope” wording. Second, sequence similarity at chromosome ends biases the homology search toward ectopic (interchromosomal) rather than purely allelic (intrachromosomal homologue-pair) repair, which is the proposed feedback-loop mechanism in C8 (chapter 07 §“Causal feedback loop”; chapter 11 finding 11; chapter 12 novel contribution #11): sequence similarity → 3D proximity at the bouquet → ectopic exchange → increased similarity. Bandyopadhyay et al. 2002’s nucleolar-association-as-cause argument for Robertsonian translocation (Bandyopadhyay et al. 2002) is the corresponding mechanism for the acrocentric C7 community: the five p-arms are co-located at a *different* nuclear landmark (the nucleolus rather than the lamina), and the spatial mechanism therefore differs from the D4Z4 / lamin tether but the principle is identical.

Hi-C, Pore-C, single-cell and meiotic 3D (topic 09)

Lieberman-Aiden et al. 2009 (Lieberman-Aiden et al. 2009) introduced Hi-C and demonstrated bulk principles of human genome folding (compartments, contact frequency vs distance, principal genome partitions); Rao et al. 2014 (Rao et al. 2014) extended this to kilobase resolution and identified looping principles. Single-cell variants followed: Nagano et al. 2013 (Nagano et al. 2013) introduced single-cell Hi-C; Stevens et al. 2017 (Stevens et al. 2017) resolved 3D structures of individual mammalian genomes; Ramani et al. 2017 (Ramani et al. 2017) built massively multiplex single-cell Hi-C; Tan, Xing and colleagues 2018 introduced Dip-C (Tan et al. 2018) and resolved GM12878 and PBMC single-cell 3D structures; Ulahannan et al. 2019 (Ulahannan et al. 2019) gave Pore-C the long-read concatemer framing.

Meiotic 3D came of age with Patel et al. 2019 (Patel et al. 2019) (mouse meiotic Hi-C, chromosome-end clustering during meiotic prophase) and especially Zuo et al. 2021 (Zuo et al. 2021) (stage-resolved mouse meiotic Hi-C across leptotene, zygotene, pachytene, diplotene; LINC complex / SUN1 force transmission; meiotic chromosome-end alignment extends roughly 20% of chromosome length; the SUN1 W151R mutant compresses this to roughly 5%). Xu et al. 2025 (Xu and others 2025) extended single-cell 3D into 20 sperm cells. For the present manuscript these papers supply the analytical backbone of C7: Mantel tests of sequence-similarity × Hi-C contact at six bulk Hi-C samples, two Pore-C and CiFi datasets, GM12878 and sperm Dip-C, and Zuo’s mouse meiotic Hi-C all converge on $\rho \approx 0.66\text{--}0.72$ (chapter 05; chapter 06; chapter 08). The mouse zygotene $\rho=0.715$ (Zuo 2021 stage-resolved Hi-C) is the strongest existing evidence that the sequence-similarity ↔ 3D-proximity correspondence peaks at the meiotic stage where the bouquet is most active.

Methodologically, three subsidiary innovations are also notable: (i) Stergachis et al. 2020 (Stergachis et al. 2020) introduced Fiber-seq for single-molecule chromatin-fibre regulatory architectures; (ii) Gershman et al. 2022 (Gershman et al. 2022) re-aligned ENCODE CTCF ChIP-seq to T2T-CHM13,

confirming CTCF enrichment at TAR loci; (iii) Tan 2018's Dip-C produced per-haplotype radial-position data that the present chapter 05 §“Nuclear lamina cross-reference” uses as an envelope proxy (C1 D4Z4 radial=0.732 peripheral; C14 PAR2 0.840). These are the analytical bricks that allow chapter 05's “the Mantel rho strengthens for ALL 8 datasets at ALL resolutions when acrocentric / sex / strongest-community exclusions are applied” — the cleanest available defence against confound-driven Hi-C signal in subtelomeres.

Synthesis of mechanism

Three mechanistic threads converge. First, *concerted-evolution machinery* (gene conversion, NAHR, BIR, segmental-duplication-mediated rearrangement) supplies the molecular events that produce sub-block similarity, with the strict-sense rDNA case (Eickbush and Eickbush 2007; Ganley and Kobayashi 2007) as the asymptote of fast intra-array homogenisation and the loose-sense subtelomere-block case as the slower inter-arm version. Second, the *meiotic bouquet* supplies a cell-biological window in which interchromosomal homology search at chromosome ends is maximal, biasing repair toward ectopic outcomes when sequence similarity exists; nucleolar association supplies an analogous window for the acrocentric short arms. Third, *Hi-C and single-cell 3D* methods supply the empirical access point: 3D contact across non-homologous chromosome ends should track sequence similarity if the bouquet hypothesis is correct, and the present manuscript reports exactly that correspondence at multiple scales. Part IV now asks how the methods (pangenomes, pedigrees) make this story testable at population scale.

Part IV — Methods: pangenomes and pedigrees (combines topics 10, 11, 12)

Pangenome graphs and IMPG transitive closure (topic 10)

The pangenome graph framework emerged from the recognition that single-reference workflows under-represent diverse alleles. The Computational Pan-Genomics Consortium 2018 (Computational Pan-Genomics Consortium 2018) consolidated the early conceptual frame; Eizenga et al. 2020 (Eizenga et al. 2020) published the canonical “Pangenome graphs” review; Garrison et al. 2018 introduced vg (Garrison et al. 2018); Li, Feng & Chu 2020 introduced minigraph (Li, Feng, and Chu 2020) for reference-based graph construction; Garrison et al. 2024 introduced pggg (pangenome graph builder) (Garrison et al. 2024) for reference-free all-vs-all graph construction; Heumos et al. 2024 published the odgi infrastructure used to compute pairwise Jaccard similarity on the pggg GFA (Heumos et al. 2024). The Computational Pan-Genomics Consortium 2018 paper (Computational Pan-Genomics Consortium 2018) (Marschall, Marz, Abeel et al.) is the consortium-level review.

For the present manuscript the canonical pipeline is: wfmask all-vs-all pairwise mapping at p95 identity over the 18,827 telomere-anchored 500 kb flanks → IMPG (<https://github.com/pangenome/impg>) transitive closure on the implicit pangenome graph → pggg GFA construction → odgi-derived pairwise Jaccard matrix → Leiden community detection at arm-level (k=15) and sequence-level (k=50). The conceptual point that the topic_10 review establishes is that the all-vs-all PAF set is the implicit pangenome graph: each PAF edge is a pairwise mapping, the union over flanks is the graph, and IMPG query -x performs transitive closure on it. The pggg-built explicit GFA is one downstream product, not the primary object. Per Erik's clarification (CROSSWALK §7a), the Methods should cite IMPG and reframe the all-vs-all PAF set as the implicit graph; the field already accepts the concept and the Methods does not need to invent it. The Erdős-Rényi connectivity argument (CROSSWALK §7b: $n=18,827 \rightarrow p^* = \log(n)/n \approx 5.21e-4$; 12% sampling is $\sim 230\times$ above threshold; the resulting random graph is densely connected and transitive closure reaches everywhere) is the formal justification for “no chromosomal partitioning.”

Three canonical companion papers extend the pangenome frame. Guarracino et al. 2023 (Guarracino et al. 2023) is the closest predecessor: pangenome-graph evidence of recombination between heterologous human acrocentric chromosomes, using HPRC v1 data. de Lima et al. 2025 (Lima, Guarracino, and others 2025) is a follow-on pangenome-resolved analysis of human Robertsonian breakpoints. Rautiainen et al. 2023 (Rautiainen et al. 2023) introduced Verkko, the diploid T2T assembly pipeline whose output feeds the WashU pedigree analysis. These three together with Liao 2023 (Liao et al. 2023) (HPRC v1) define the pangenome-method branch of the lineage that the present manuscript continues.

Pedigree-based variant and recombination detection (topic 11)

Topic 11 supplies the second methodological branch. The committed topic_11 review [paper_prep/lit_review/topic_11_pedigree_based_recombination_detection.md] documents this in detail. Briefly: deCODE's fine-scale pedigree maps (Kong et al. 2010; Bhérer, Campbell, and Auton 2017) and Halldorsson et al.'s sequence-level genetic map (Halldórsson et al. 2019) established the logic of using transmitted alleles to infer recombination and de novo mutation; Sasani et al. 2019's three-generation Utah families (Sasani et al. 2019) showed how grandchildren resolve postzygotic vs germline mosaicism; Chaisson et al. 2019 (Chaisson et al. 2019), Audano et al. 2019 (Audano et al. 2019), Collins et al. 2020 (Collins et al. 2020) and Ebert et al. 2021 (Ebert et al. 2021) showed that long-read or multi-platform structural-variation discovery exposed many variants invisible to short reads; Cheng et al. 2021 (hifiasm) (Cheng et al. 2021) and Rautiainen et al. 2023 (Verkko) (Rautiainen et al. 2023) supplied haplotype-resolved assembly methods; Wagner et al. 2022 (Wagner et al. 2022) and Zook et al. 2020 (Zook et al. 2020) gave benchmarking frameworks for difficult variants and SVs; Ahsan et al. 2023 (Ahsan et al. 2023) and Liu et al. 2024 (Liu et al. 2024) reviewed long-read SV detection algorithm tradeoffs; Nurk et al. 2022 (Nurk et al. 2022) and Liao et al. 2023 (Liao et al. 2023) supplied the T2T-CHM13 reference and HPRC v1 pangenome.

The two papers that are most directly load-bearing for C8 are Cechova et al. 2025 (Cechova and others 2025) and Porubsky et al. 2025 (Porubsky and others 2025). Cechova et al. 2025 supplies the WashU 3-generation T2T pedigree (PAN010/PAN011 → PAN027 → PAN028), the primary substrate for the abstract's "538 patches of recent recombination, 92% confined within sequence-similarity communities" claim. Porubsky et al. 2025 supplies the CEPH1463 4-generation pedigree (28 hifiasm + 14 verkko phased assemblies), used for cross-assembler validation: 11 robust inter-chromosomal subtelomeric features that replicate across both pipelines (chr10/chr18 in NA12877+NA12878 = the Linardopoulou 2005 pair; chr19/chr22 transmitted via NA12878; chr12/chr9 in both NA12889 and NA12890). The methodological principle the topic_11 review documents most carefully — and that the manuscript's chapter 14 inherits — is that an inter-chromosomal patch in a child is evidentially *strongest* when it (a) sits on an inherited parental haplotype with the expected Mendelian phase, (b) is reproducible across independent assemblers or generations, and (c) sits inside a Leiden community independently defined from unrelated HPRC v2 haplotypes.

HPRC v1 / v2 population pangenomes (topic 12)

The HPRC literature is the dataset substrate. Nurk 2022 (Nurk et al. 2022) (T2T-CHM13) closed the reference-genome side; Liao 2023 (Liao et al. 2023) is the HPRC v1 draft pangenome (47 individuals, 94 haplotypes); HPRC v2 (forthcoming, preprinting per lead-author clarification) extends this to 233 individuals × 2 haplotypes = 465 plus CHM13 = 466 haplotype-equivalent units in the present manuscript. The technical story is straightforward: hifiasm (Cheng et al. 2021) and Verkko (Rautiainen et al. 2023) deliver haplotype-resolved assemblies; pggb (Garrison et al. 2024) / odgi (Heumos et al. 2024) / IMPG (in submission) deliver the pangenome-graph operations; Wagner 2022 (Wagner et al. 2022) / Zook 2020 (Zook et al. 2020) / GIAB-style benchmarking discipline keeps the dataset rigorous. The dataset is large enough that subtelomeric block frequencies become

measurable as population variables; this is the single technical advance that makes C5 (cladistic structure with named clades) and C6 (population clustering across human ancestries) possible.

The methodological hand-off from topic 10 → topic 11 → topic 12 is the architecture of the present paper's Methods: pangenome graph ingests 466 haplotypes; community detection partitions them; pedigree analysis tests whether the partition predicts inheritance; population-genetics analysis tests whether the partition tracks ancestry. Each branch supplies a different *kind* of validation. The pangenome graph supplies *internal* structure (sequence-similarity communities). The pedigree supplies *transmission* validation (do new generations inherit and recombine within those communities?). The population-genetics analysis supplies *external* validation (do those communities mirror known population structure?). These three together are the empirical core of C5–C8.

Part V — How this paper sits in the field

Part V is the synthesis the manuscript itself depends on: which of the abstract's eight claims are old discoveries reproduced at scale, which are new, and which depend on framing decisions that the abstract has chosen to make explicit. This section follows the C-numbering of `paper_prep/synthesis/CROSSWALK.md`.

What is recovered (pre-existing literature confirmed at population scale)

C1 (HPRC v2 companion-paper framing) is editorial rather than empirical; it commits the present manuscript to the HPRC v2 publication arc and is correct as soon as the HPRC v2 main paper is in press. C2 (implicit pangenome graph; ~12% pairwise sampling) reframes the standard wfmmash + pggg + odgi pipeline as a graph object and supplies an Erdős-Rényi connectivity argument; the *concept* is established in (Eizenga et al. 2020; Computational Pan-Genomics Consortium 2018) and the *implementation* is established in (Garrison et al. 2018; 2024; Heumos et al. 2024). C3 (466 near-complete haplotype assemblies) is direct continuation of (Liao et al. 2023) using HPRC v2; the dataset itself is the new contribution.

C4 (extended interchromosomal homology at nearly all subtelomeres comparable to PAR2) recovers and dramatically extends Flint, Bates et al. 1997 (Flint et al. 1997) and Riethman et al. 2004 (Riethman et al. 2004). The two-domain model is confirmed on 39/48 arms by gradient analysis and 39/41 by piecewise breakpoint analysis (chapter 04 §“Two-domain subtelomeric model test”); internal (TTAGGG)_n islands co-localise with the proximal-domain boundary within 25 kb on 11/19 testable arms (chapter 04). The “comparable to PAR2” wording is a *new* explicit comparison; the Flint era did not run this comparison directly.

C5 (cladistic analysis with named clades) recovers Linardopoulou 2005 (Linardopoulou et al. 2005) (chr10p–chr18p), Mefford & Trask 2002 (Mefford and Trask 2002) (f7501-cluster, chr4q–chr10q D4Z4), Lemmers 2010 (Lemmers et al. 2010) (chr4qA / DUX4 polymorphism), and Trask 1998 (Trask et al. 1998) (OR cluster polymorphism), and supplies a new {22q, 21q, 19q, 1q, 13q, 17q} clade plus a large moderate-similarity clade. Note framing: the abstract names “neighbour-joining trees” but Andrea's analysis runs Leiden + UPGMA with 12-of-15-community agreement (chapter 01); the named clades all map to communities, so this is editorial rather than analytical. The novel contribution is *quantitative* (population-scale community structure of 41 arms across 466 haplotypes — chapter 12 novel contribution #1) and *cladistic resolution* (named clades that the reference-only era could not assign confidently).

C6 (PCA / community detection on similarity matrix → subtelomere clustering across human populations) recovers Mefford & Trask 2002's hedge about whether subtelomeric block frequencies could be phylogenetic markers. The answer is positive: the out-of-Africa Fst tree from chapter 04 /

chapter 12 novel contribution #19 recovers AFR as the deepest split with mean F_{st} 0.044, AFR/non-AFR F_{st} 0.10–0.15 vs non-AFR 0.00–0.02; Mefford’s hedge is now a confirmed positive result.

C7 (Hi-C 3D maps testing nuclear-envelope-proximity recombination hypothesis) recovers Stout et al. 1999 (Stout et al. 1999) (interphase chr4q/chr10q clustering by FISH), Masny 2004 (Masny et al. 2004) (4q35.2 peripheral lamin-A/C tethering), Ottaviani 2009 (Ottaviani et al. 2009b) (D4Z4-CTCF-lamin), Zuo 2021 (Zuo et al. 2021) (chromosome-end alignment extends ~20% in mouse meiosis), Patel 2019 (Patel et al. 2019) (mouse meiotic Hi-C), Acquaviva 2020 (Acquaviva et al. 2020) (PAR-axis-elongation force on DSB localisation in mouse), and Chu 2017 (Chu et al. 2017) (PAR-TERRA-mediated sex-chromosome pairing). The novel contribution is *direct quantitative testing* in human bulk and single-cell Hi-C and Pore-C: Mantel $\rho \approx 0.66$ across 8 datasets, strengthening to 0.79 after exclusion controls (chapter 05). The “flanking-paradox” finding (chapter 05 / chapter 07 / chapter 11 finding 9) — unique-sequence flanking shows *stronger* 3D enrichment than duplicated PHR — rules out multi-mapping artifacts. The framing gap noted in CROSSWALK §6 is that no direct LAD / Lamin B1 ChIP overlay exists yet; Dip-C radial position is used as a proxy. REWRITE_PLAN TASK-22 is the appropriate follow-up; if unavailable, the abstract should soften “nuclear envelope” to “peripheral telomere positioning.”

C8 (synthesising thesis: “concerted evolution and unorthodox recombination”) recovers Dover 1982 (Dover 1982) / Liao 1999 (Liao 1999) (concerted evolution as molecular drive), Stankiewicz & Lupski (Stankiewicz and Lupski 2002) (NAHR), Henderson 1972 (Henderson, Warburton, and Atwood 1972) / Floutsakou 2013 (Floutsakou et al. 2013) / McStay 2016 (McStay 2016) (rDNA-flanking concerted evolution), Linardopoulou 2005 (Linardopoulou et al. 2005) (subtelomeres as hot spots of interchromosomal recombination), and Guarracino 2023 (Guarracino et al. 2023) (pangenome-graph evidence for inter-acrocentric recombination); the v3-opus topic_05 refresh explicitly adds de Lima 2025 (Lima, Guarracino, and others 2025) (SST1 macrosatellite as the recombining homology for dominant Robertsonian fusions) and Stults 2008 (Stults et al. 2008) (>10% per-meiosis rDNA-cluster rearrangement frequency) as direct mechanistic anchors for the loose-sense usage. The single most-novel contribution is **direct pedigree evidence**: 538 high-quality interchromosomal patches in the WashU T2T pedigree, 92% within Leiden communities, including 133 gene-conversion-like events at perfect 1.000/1.000 score, 16 crossover-like events, 229 acro-like events; the chr10q $\leftarrow$ chr4q gene-conversion event observed at score 0.957 in PAN028 maternal. Per the lead-author clarification (CROSSWALK §5), the pedigree IS the proof of concerted evolution in the loose sense.

What is genuinely new

A short list, in priority order:

1. **Direct pedigree evidence of ongoing inter-chromosomal exchange** (chapter 14). 538 patches, 92% within communities, 133 gene-conversion-like at perfect score, 16 crossover-like, transmitted across three generations in WashU T2T plus 11 cross-assembler-validated CEPH1463 features. This is the most-novel contribution and the empirical core of C8.
2. **Population-scale community structure** (chapter 01). 41 arms in 15 arm-level Leiden communities and 50 sequence-level communities across 466 haplotypes, the first population-scale quantification of the patchwork model.
3. **Two-domain Flint/Mefford model confirmed at pangenome scale** (chapter 04). 39/48 arms gradient, 39/41 piecewise, 11/19 ITS co-localised, 99.7% per-haplotype. Extends Flint 1997 by a factor of ~13.

4. **Subtelomeric Fst mirrors out-of-Africa** (chapter 04 / chapter 12 novel contribution #19). Subtelomeric block frequencies do carry phylogenetic signal; Mefford & Trask's 2002 hedge is now a positive result.
5. **3D-mirrors-sequence at multiple scales and resolutions** (chapter 05 / chapter 06 / chapter 07 / chapter 08). Mantel $\rho \approx 0.66$ – 0.79 across 8 bulk Hi-C / Pore-C / CiFi / Dip-C / sperm / mouse meiotic Hi-C datasets, with mouse zygotene $\rho=0.715$.
6. **Allele-vs-paralog Wilcoxon at population scale** (chapter 04). 5,946 paired distances, 8/9 multi-arm communities allele-closer (overall $p < 1e-300$); C7 acrocentric reverses to paralog-closer ($p=2.0e-7$) as quantitative confirmation of complete homogenisation.
7. **Cross-arm type discordance up to 47.5%** at chr22q (chapter 04). Refines Mefford 2002's chr4q/chr10q figure from ~20% to 43.4%.
8. **Flanking-paradox** (chapter 05 / chapter 07 / chapter 11 finding 9). Unique-sequence 100 kb regions centromere-ward of PHR boundaries show *stronger* 3D enrichment than the duplicated PHR itself — a strong methodological defence against multi-mapping artifacts.
9. **Three-category arm classification** (chapter 04 / chapter 11 finding 2). Homogeneous (8/41 with 0% cross-arm), polymorphic (34/41), fully interchangeable (acrocentric p-arms). First quantitative framework extending Mefford's qualitative patchwork model.
10. **Gene-replacement scoring** (chapter 04). Complete (0.91–1.0) at chr13_p/chr14_p/chr15_p and PAR; partial (0.0–0.72) at autosomal communities. First quantitative gradient of homogenisation onto community structure.
11. **OR4F pseudogenisation gradient** (chapter 03). 62.1% pseudogenes across 5,023 OR4F annotations; pseudogenisation rate 11.1%–99.8% across 16 arms.

Framing decisions explicit in the abstract

Three framing decisions matter for Q&A:

- **“Implicit pangenome graph” terminology and “~12% pairwise sampling”** (C2): the standard pipeline is wfmask all-vs-all + pggb + odgi; the abstract reframes the all-vs-all PAF set as the implicit graph, citing IMPG, and supplies an Erdős-Rényi argument for why no chromosomal partitioning is required. Per CROSSWALK §6 modify-task TASK-10 / TASK-11, this is editorial rather than analytical.
- **“Neighbour-joining trees”** (C5): the analysis runs Leiden + UPGMA, which agree on 12 of 15 communities. The named clades all map; the algorithm differs. Either run NJ on the existing arm-distance matrix (a one-line `ape::nj()` call) or relax the abstract to “cladistic analysis (Leiden on the arm-level Jaccard distance)”. Per Erik's clarification, both are legitimate.
- **“Concerted evolution”** (C8 title term): used in the *loose* sense (ongoing recombination exchange producing similarity homogenisation), not the strict molecular-evolution sense. Pedigree IS the evidence. The Discussion paragraph should make this explicit and reference chapter 14. Nei & Rooney 2005 (Nei and Rooney 2005) is the explicit alternative model that justifies the loose-sense framing — not all subtelomeric paralog families behave like rDNA.

A fourth, less-discussed framing decision is the **“facilitated by ... nuclear envelope”** wording (C7). Andrea's chapter 05 §“Nuclear lamina cross-reference” uses Dip-C radial position as a proxy; no genome-wide LAD / Lamin B1 ChIP overlay exists. The abstract should either be supplemented with a LAD overlay (REWRITE_PLAN TASK-22) or softened to “consistent with peripheral telomere positioning” / “consistent with envelope association.”

Closing

The Guarracino & Garrison subtelomere manuscript stands at the convergence of three decades of subtelomeric biology, six layers of mechanism (concerted-evolution machinery, NAHR, gene conversion, BIR, segmental duplication, the meiotic bouquet, pangenome representation), and two new empirical access points (HPRC v2 466-haplotype pangenome graph; T2T multigenerational pedigrees). Its single most-novel contribution is the pedigree evidence in chapter 14: subtelomeric exchange among non-homologous chromosome ends is *currently happening* in real human families, and it happens predominantly within sequence-similarity communities that were defined independently from unrelated population samples. That is the empirical proof of “ongoing recombination” / “concerted evolution” in the abstract title. Everything else in the manuscript — the pangenome graph, the cladistic structure, the Hi-C 3D evidence, the population-genetics signatures — is corroborating evidence at population scale for what the pedigree directly observes at family scale. Read in this order, the field history makes the manuscript’s claims feel inevitable rather than surprising.

The v2 refresh deepens this story in two specific places. Topic 05’s v3-opus pass adds Krystal 1981, Schmickel 1973, Worton 1988, Bandyopadhyay 2001/2002, Stults 2008, Stimpson 2014 and the de Lima 2025 SST1 mechanism — converting the acrocentric clade from a recovered cytogenetic clade to a sequence-resolved, mechanism-anchored community with a per-meiosis rate (>10%, Stults 2008) that is the highest documented in human structural biology. Topic 07’s v3-opus pass adds Brown 1972, Lupski 1998, Lupski & Stankiewicz 2005, Bailey & Eichler 2006, Eickbush & Eickbush 2007, Ganley & Kobayashi 2007, Llorente 2008, Hastings 2009, Anand 2013, Williams 2015 and the explicit Nei & Rooney 2005 birth-and-death alternative — converting the C8 mechanism story from a single-vocabulary NAHR claim to a layered framework (NAHR + gene conversion + BIR + gBGC, with birth-and-death as the explicit alternative for OR4F-like families). Both upgrades make the manuscript’s loose-sense “concerted evolution” usage defensible against a strict-sense critic.

End of SYNTHESIS_v2.md.

Appendix A — Chronology of Subtelomere Research

Author: lit-review-v2 (agent-937), 2026-05-06. Supersedes CHRONOLOGY.md (lit-review-synthesis, agent-877). **Source:** distilled from paper_prep/synthesis/REFERENCES_v3.bib (295 entries) and end-to-end-report/report/12_literature.md. Each row corresponds to one published paper. Claims supported (C1–C8) follow paper_prep/synthesis/CROSSWALK.md. Brief description distinguishes mechanism, dataset, method, prediction, or framework. Entries are sorted chronologically; ties broken alphabetically by first-author surname.

Versioning note: v2 supersedes the v1 chronology and reflects the final REFERENCES_v3.bib merge that incorporates the v3-opus refresh of topic 05 (acrocentric / rDNA / Robertsonian) and topic 07 (concerted evolution / NAHR / gene conversion).

Year	Author(s)	Brief description	Claim(s)	Citation key
1916	Robertson	Robertsonian acrocentric–acrocentric fusion concept (grasshoppers); eponym for human acrocentric short-arm rearrangements.	C5	Robertson1916
1916	Robertson	XII.—Chromosome studies. I. Taxonomic relationships shown in the chromosomes of Tettigidae and Acrididae: V...	cytofound_Robertson	Robertson1916
1970	Caspersson et al.	Differential binding of alkylating fluorochromes in human chromosomes	cytofound_Caspersson	Caspersson1970
1970	Caspersson et al.	Quinacrine chromosome banding; first method to identify all 24 human chromosomes; G-banding nomenclature.	C1 C5	Caspersson1970
1970	Caspersson et al.	The 24 fluorescence patterns of the human metaphase chromosomes — distinguishing characters and variability	cytofound_Caspersson	Caspersson1971
1972	Brown et al.	Xenopus rDNA homogenisation; empirical pattern that motivated Smith 1976's unequal-crossover model.	C8	Brown1972
1972	Henderson et al.	In situ localisation of rDNA arrays to all five acrocentric short arms; original cross-chromosome concerted-evolution observation.	C5 C8	Henderson1972
1972	Lewontin	The Apportionment of Human Diversity	subtelopopgen_lewontin	lewontin1972
1975	Schmickel	Quantitation of human ribosomal DNA: hybridization of human DNA with ribosomal RNA for quantitation and fra...	acrocentric_rDNA_robertsonian_schmickel	Schmickel1975
1976	Smith	Unequal crossover among tandem repeats produces homogenisation; mechanistic basis of concerted evolution.	C7, C8	Smith1976
1978	Blackburn & Gall	Original (TTAGGG) _n -like canonical telomere repeat in Tetrahymena rDNA termini; molecular target of TAR1 and ITS biology.	C1 C5	BlackburnGall1978
1979	Hsu	Standard historical reference for cytogenetic foundations (banding, FISH, satellite-DNA cytology).	C5	Hsu1979
1980	Arnheim et al.	Molecular evidence for genetic exchanges among ribosomal genes on nonhomologous chromosomes in man and apes	concerted_evolution_nahr_Arnheim	Arnheim1980
1981	Krystal et al.	Human nucleolus organizers on nonhomologous chromosomes share the same ribosomal gene variants	acrocentric_rDNA_robertsonian_krystal	Krystal1981
1982	Burgoyne	Obligate X/Y crossover at PAR1; founding statement of pseudoautosomal recombination.	C4, C7	Burgoyne1982
1982	Dover	Concerted evolution / molecular drive theory; foundational framework for repetitive-sequence homogenisation.	C8	Dover1982
1983	Guichaoua et al.	Structural basis for Robertson translocations in man: association of acrocentric ribosomal genes in the nucleolar fibri...	C5 C8	Guichaoua1983
1984	Ohta	Some models of gene conversion for treating the evolution of multigene families	concerted_evolution_nahr_Ohta	Ohta1984
1984	Weir & Cockerham	Estimating F-statistics for the analysis of population structure	subtelopopgen_weir	Weir1984
1985	Greider & Blackburn	Identification of a specific telomere terminal transferase activity in Tetrahymena extracts	cytofound_Greider	Greider1985
1986	Dover	Molecular drive: a third mode of evolution alongside selection and drift	C8	Dover1986
1987	Smith	Unequal crossover among tandem repeats produces homogenisation; mechanistic basis of concerted evolution.	C7, C8	Smith1987

Total: 295 entries.

End of CHRONOLOGY_v2.md.

Appendix B — Open Questions and Gaps

Author: lit-review-v2 (agent-937), 2026-05-06. Supersedes GAPS.md (lit-review-synthesis, agent-877).

Companion files: SYNTHESIS_v2.md, CHRONOLOGY_v2.md, paper_prep/synthesis/REFERENCES_v3.bib, paper_prep/synthesis/CROSSWALK.md.

This is a one-page note flagging anything the v2 synthesis pass surfaced as missing, mis-cited, or open. It is not a follow-up plan; only a list of items that future passes (or revisions to the manuscript) should resolve. The v1 GAPS.md is retained as historical record; almost all of its entries carry forward unchanged.

Substrate provenance

All 14 topical reviews are committed under paper_prep/lit_review/: topic_01 cytogenetic foundations (21 bib entries), topic_02 TAR1 / TTAGGG / ITS structure (16), topic_03 pseudohomologous regions concept (14), topic_04 sex-chromosome pseudoautosomal regions (28), topic_05 acrocentric / rDNA / Robertsonian (22 — v3-opus refresh), topic_06 D4Z4 / DUX4 / FSHD (13), topic_07 concerted evolution / NAHR / gene conversion (24 — v3-opus refresh), topic_08 meiotic bouquet / nuclear-envelope tethering (19), topic_09 Hi-C / Pore-C / single-cell / meiotic 3D (16), topic_10 pangenome graphs / IMPG (15), topic_11 pedigree-based recombination detection (20), topic_12 HPRC v1 / v2 population pangenomes (17), topic_13 subtelomere population genetics / Fst / out-of-Africa (12), topic_14 OR / OR4F gradient (17). Total raw substrate: 254 entries across the 14 topic bibs, with 24 in the original REFERENCES.bib.

REFERENCES_v3.bib merges REFERENCES.bib (24), the prior synth-pass merge REFERENCES_v2.bib (270; carried forward for canonical-key continuity), and all 14 topic_NN*.bib (254 raw, 240 unique after intra-topic dedup) into 295 deduplicated entries. Dedup is by DOI (when present) and by (first-author surname, year, normalized-title prefix) with the canonical / prefix-free key preferred on collision so existing SYNTHESIS_v2.md citations resolve unchanged. All 168 distinct citations in SYNTHESIS_v2.md resolve in REFERENCES_v3.bib (verified 2026-05-06).

The substantive change from v1 to v2 is the v3-opus refresh of two topical reviews: - **topic_05** grew from ~16 entries to 22, adding Schmickel 1973, Krystal 1981, Worton 1988, Page 1996, Bandyopadhyay 2001/2002, Stults 2008, Stimpson 2014, Cechova/Hartley 2025 (great-ape acrocentric atlas) and Hartley 2026 (UK Biobank ROB-carrier genotyping). - **topic_07** grew from ~18 entries to 24, adding Brown 1972, Lupski 1998, Lupski & Stankiewicz 2005, Bailey & Eichler 2006, Eickbush & Eickbush 2007, Ganley & Kobayashi 2007, Llorente 2008, Hastings 2009, Anand 2013, Williams 2015, Logsdon 2024 and the explicit Nei & Rooney 2005 birth-and-death alternative.

Both refreshes are integrated into SYNTHESIS_v2.md Parts II (acrocentric/rDNA section) and III (concerted evolution / NAHR section). The closing paragraph of SYNTHESIS_v2.md Part V calls out the specific upgrades.

Bibliography-level gaps (mostly carried forward from v1)

- **Cechova2025** is cited as “(in press)” with no DOI; the WashU pedigree paper is not yet on bioRxiv per the lead author’s clarification. Update DOI once available.
- **Porubsky2025** is cited as “(in press)” with no DOI; the CEPH1463 paper was published in *Nature* (April 2025) per the lead author’s recent comment. Update with the published DOI before submission.
- **deLima2025** is “(in submission)” ; verify status before submission. (Note: topic_05 v3-opus refresh assigns this paper a sequence-resolved SST1-mechanism finding that is now load-bearing for C8.)
- **Xu2025** lists “Hanbo Xu and others” — replace with full author list once the manuscript is in its published form.

- **Francis2025** has full DOI (10.1038/s41588-025-02358-1) but author list is currently truncated to “Brittany A. Francis and others” — replace with full author list before submission.
- **Stergachis Fiber-seq telomere companion paper.** REFERENCES_v3.bib cites Stergachis2020 with a note that the specific Fiber-seq-at-39-of-46-telomeres paper is the relevant downstream work — track that paper down before submission.
- **StankiewiczLupski2002 and StankiewiczLupski2010** are both cited; verify the 2002 *Trends in Genetics* paper and the 2010 *Annual Review of Medicine* paper resolve to the intended NAHR vocabulary in context.
- **Computational2018 (consortium-author key)** is the canonical citation for the *Computational pan-genomics* review (Marschall, Marz, Abeel et al., *Briefings in Bioinformatics* 2018, doi:10.1093/bib/bbw089). Duplicate author-led keys are removed in REFERENCES_v3.bib.
- **Topic-prefix vs canonical-key duplication.** Several topical reviews use prefixed keys (e.g. sexchrompars_charchar2003, acrocentric_rdna_robertsonian_henderson1972, dux4_d4z4_fshd_lemmers2010) for papers that the existing REFERENCES.bib / prior synthesis pass cite as canonical PascalCase (Charchar2003, Henderson1972, Lemmers2010). The merge into REFERENCES_v3.bib keeps the canonical key on each duplicate (per the dedup rule: prefer prefix-free) so SYNTHESIS_v2.md and CHRONOLOGY_v2.md citations continue to resolve. The topic_NN_*.md files (read-only) still use their prefix keys and must therefore be compiled against topic_NN_*.bib directly, not against REFERENCES_v3.bib alone. The unique-to-topic prefix keys survive in REFERENCES_v3.bib unchanged.
- **One renamed citation in v2.** SYNTHESIS_v2.md cites @ChenCooper2007 rather than the v1 @Chen2007; the entry is the same paper (Chen, Cooper, Krawczak, *Hum. Mutat.* 2007 gene-conversion review). The v3 dedup preferred the longer ChenCooper2007 key on the (Chen, 2007, gene-conversion) collision because both keys are non-prefixed.

Substantive gaps

- **C7 — direct LAD / Lamin B1 ChIP overlay.** The abstract’s “facilitated by the physical proximity of subtelomeres at the nuclear envelope” wording rests on Dip-C radial position as a proxy and on the Masny 2004 / Ottaviani 2009 mechanism for D4Z4 — chapter 05 §“Nuclear lamina cross-reference” does not run a genome-wide LAD intersection. REWRITE_PLAN.md TASK-22 is the correct follow-up. Fallback per CROSSWALK §6: soften to “consistent with peripheral telomere positioning.”
- **C5 — neighbour-joining tree** is named in the abstract but Andrea uses Leiden + UPGMA (12 of 15 communities agree exactly). One-line fix is ape: :nj () on the existing arm-distance matrix; alternatively, relax the abstract wording to “cladistic analysis (Leiden / UPGMA).” Per REWRITE_PLAN.md TASK-01 / TASK-13.
- **C2 — “~12% pairwise sampling” computation.** Methods must compute the wfmash k-mer-evaluation rate from the on-disk PAFs (the realised value, not asserted) and write the Erdős-Rényi argument explicitly ($n=18,827$; $p^* = \log(n)/n \approx 5.21e-4$; 12% is $\sim 230\times$ above). Per REWRITE_PLAN.md TASK-10 / TASK-11. The argument is in CROSSWALK §7b.
- **C2 / C3 dataset count.** Andrea reports 465 throughout; the abstract reports 466. Resolve canonically as $466 = 233 \text{ individuals} \times 2 \text{ haplotypes} + \text{CHM13 reference}$ (per CROSSWALK §7c).
- **C8 — “concerted evolution” loose-sense framing.** The Discussion paragraph should explicitly state the loose-sense usage and tie it to the pedigree (chapter 14) as the empirical anchor. Per CROSSWALK §3 and REWRITE_PLAN.md TASK-19. The v3-opus topic_07 refresh adds Nei & Rooney 2005 (Nei and Rooney 2005) as the explicit alternative model: not all subtelomeric paralog families behave like rDNA, and the loose-sense framing is what allows the abstract to encompass both homogenising (rDNA) and turnover-prone (OR4F) systems.

New gaps surfaced by the v2 substrate

- **C5 — acrocentric per-meiosis rate.** Stults et al. 2008 (Stults et al. 2008) reported >10% rDNA-cluster rearrangement per meiosis using optical pulsed-field gel electrophoresis. Andrea's chapter 14 reports 133 gene-conversion-like patches across the WashU pedigree but does not separate intra-NOR conversion from inter-arm transfer at sequence resolution. A direct comparison of Stults' bulk rate vs the pedigree's per-arm rate would either confirm or revise the >10% number — useful for the Discussion.
- **C5 — rare Robertsonian products (rob(13;15), rob(14;15), rob(15;22)) lack sequence-resolved breakpoints.** de Lima/Guarracino 2025 (Lima, Guarracino, and others 2025) resolved rob(13q14q) and rob(14q21q) via the SST1 mechanism. Whether the rarer products use the same SST1 mechanism or a different homology block (rDNA itself, satellite III subfamilies, distal junction) is open; the Hartley 2026 (Hartley et al. 2026) UK Biobank cohort may eventually carry enough rare carriers to address this.
- **C8 — relative contribution of NAHR vs gene conversion vs BIR.** The 133 gene-conversion-like + 16 crossover-like + 229 across-like patches in chapter 14 imply gene conversion dominates by an order of magnitude, but BIR-mediated tracts of 100+ kb may be miscategorised as gene conversion under the current sandwich-pattern heuristic. Anand et al. 2013 (Anand, Lovett, and Haber 2013) flags exactly this miscategorisation risk in yeast. A future pass could add a tract-length-aware classifier that distinguishes classical gene conversion (50–500 bp, per Jeffreys & May 2004 (Jeffreys and May 2004)) from BIR-style long tracts (10–100+ kb, per Llorente 2008 (Llorente, Smith, and Symington 2008)).
- **C8 — gBGC fingerprint at PHR boundaries.** Galtier 2001 (Galtier et al. 2001) / Duret & Galtier 2009 (Duret and Galtier 2009) establish that gene conversion is GC-biased. A per-PHR-boundary GC-content analysis on the existing pangenome graph is testable from current data and would either confirm or reject gBGC as a contributor to subtelomeric base composition.

Open empirical questions surfaced by the synthesis (carried forward)

- **Crossover-rate ↔ cross-arm-affinity correlation honest negative.** End-to-end-report chapter 12 testable prediction #7 reports $\rho = -0.43$, $p = 0.006$ across all 39 arms but $\rho = 0.00$ after excluding 7 short-read-confounded arms. Open question: does a higher-quality recombination map (deeper sequencing, longer reads at acrocentric / PAR arms) recover the predicted negative correlation? Probably yes, but cannot be answered without new data.
- **LINC-complex / SUN1-mutant test of meiotic alignment at PHR scale.** Zuo 2021 wild-type vs SUN1-W151R zygotene Hi-C is available (GEO: GSE155142, GSE155638, GSE155967). The PHR-median 105 kb scale is well below the ~5% mutant tip zone; tip contacts may be maintained without LINC-mediated force transmission, so the predicted effect is uncertain. Per chapter 12 testable prediction #1.
- **Somatic vs germline exchange.** HPRC v2 sample-source metadata (LCL vs blood) is not currently available; the manuscript cannot directly distinguish meiotic from somatic-LCL exchange in chapter 14 patches. The pedigree's *inheritance* across three generations does rule out purely somatic origin for the transmitted patches but does not directly establish meiotic timing in cells.
- **Subtelomeric Fiber-seq / nucleosome footprint at PHR boundaries.** The Stergachis 2020 framework supports single-molecule CTCF / nucleosome footprinting; a per-PHR-boundary analysis at 39 / 46 telomeres would either confirm or reject the CTCF-cohesin boundary prediction (chapter 12 testable prediction #4). Not currently in scope but a clean follow-up.

What this synthesis does *not* claim

This document does not extend any of the abstract's claims beyond what paper_prep/synthesis/CROSSWALK.md, paper_prep/synthesis/REFERENCES.bib, the 14 committed topical reviews (topic_01_* through topic_14_*), the v3-opus topic_05 and topic_07 refreshes, and end-to-end-report/report/12_literature.md already establish. The prose in SYNTHESIS_v2.md operates at the field-history-and-framework level; where deeper per-paper analysis is needed, the corresponding topic_NN_*.md provides it. No new analytical claim is made; this is a literature synthesis only.

End of GAPS_v2.md.

References

- 1000 Genomes Project Consortium. 2015. "A Global Reference for Human Genetic Variation". *Nature* 526 (7571): 68–74. <https://doi.org/10.1038/nature15393>.
- Acquaviva, Laurent, Michiel Boekhout, Mehmet E. Karasu, Kevin Brick, Florencia Pratto, Tao Li, Megan van Overbeek, et al. 2020. "Ensuring Meiotic DNA Break Formation in the Mouse Pseudoautosomal Region". *Nature* 582:426–31. <https://doi.org/10.1038/s41586-020-2327-4>.
- Adam, Eleni, Desh Ranjan, and Harold Riethman. 2022. "NPGREAT: Assembly of Human Subtelomere Regions with the Use of Ultralong Nanopore Reads and Linked-Reads". *BMC Bioinformatics* 23:545. <https://doi.org/10.1186/s12859-022-05081-3>.
- Ahsan, Mian Umair, Qian Liu, Jonathan Elliot Perdomo, Li Fang, and Kai Wang. 2023. "A Survey of Algorithms for the Detection of Genomic Structural Variants from Long-Read Sequencing Data". *Nature Methods* 20:1143–58. <https://doi.org/10.1038/s41592-023-01932-w>.
- Ambrosini, Anthony, Sheila Paul, Sufeng Hu, and Harold Riethman. 2007. "Human Subtelomeric Duplicon Structure and Organization". *Genome Biology* 8 (7): R151. <https://doi.org/10.1186/gb-2007-8-7-r151>.
- Anand, Ranjith P., Susan T. Lovett, and James E. Haber. 2013. "Break-Induced DNA Replication". *Cold Spring Harbor Perspectives in Biology* 5 (12): a10397. <https://doi.org/10.1101/cshperspect.a010397>.
- Audano, Peter A., Arvis Sulovari, Tina A. Graves-Lindsay, Stuart Cantsilieris, Melanie Sorensen, Anne E. Welch, Mark L. Dougherty, et al. 2019. "Characterizing the Major Structural Variant Alleles of the Human Genome". *Cell* 176 (3): 663–75. <https://doi.org/10.1016/j.cell.2018.12.019>.
- Bailey, Jeffrey A., and Evan E. Eichler. 2006. "Primate Segmental Duplications: Crucibles of Evolution, Diversity and Disease". *Nature Reviews Genetics* 7 (7): 552–64. <https://doi.org/10.1038/nrg1895>.
- Bailey, Jeffrey A., Zhiping Gu, Royden A. Clark, Knut Reinert, Rhea V. Samonte, Stuart Schwartz, Mark D. Adams, Eugene W. Myers, Peter W. Li, and Evan E. Eichler. 2002. "Recent Segmental Duplications in the Human Genome". *Science* 297 (5583): 1003–7. <https://doi.org/10.1126/science.1072047>.
- Bandyopadhyay, R., A. Heller, C. Knox-DuBois, C. McCaskill, S. A. Berend, S. L. Page, and L. G. Shaffer. 2002. "Parental Origin and Timing of De Novo Robertsonian Translocation Formation". *American Journal of Human Genetics* 71 (6): 1456–62. <https://doi.org/10.1086/344662>.
- Bandyopadhyay, R., C. McQuillan, S. L. Page, K. H. A. Choo, and L. G. Shaffer. 2001. "Identification and Characterization of Satellite III Subfamilies to the Acrocentric Chromosomes". *Chromosome Research* 9 (3): 223–33. <https://doi.org/10.1023/A:1016648404388>.
- Bellott, Daniel W., Jennifer F. Hughes, Helen Skaletsky, Erik C. Owen, and David C. Page. 2024. "Where Is the Boundary of the Human Pseudoautosomal Region?". *American Journal of Human Genetics* 111 (11): 2530–41. <https://doi.org/10.1016/j.ajhg.2024.09.005>.
- Bergman, Juraj, and Mikkel Heide Schierup. 2022. "Evolutionary Dynamics of Pseudoautosomal Region 1 in Humans and Great Apes". *Genome Biology* 23:215. <https://doi.org/10.1186/s13059-022-02784-x>.
- Bergström, Anders, Shane A. McCarthy, Ruoyun Hui, Mohamed A. Almarri, Qasim Ayub, Petr Danecek, Yuan Chen, et al. 2020. "Insights into Human Genetic Variation and Population History from 929 Diverse Genomes". *Science* 367 (6484): eaay5012. <https://doi.org/10.1126/science.aay5012>.

- Bhérier, Claude, Christopher L. Campbell, and Adam Auton. 2017. "Refined Genetic Maps Reveal Sexual Dimorphism in Human Meiotic Recombination at Multiple Scales". *Nature Communications* 8:14994. <https://doi.org/10.1038/ncomms14994>.
- Blackburn, Elizabeth H., and Joseph G. Gall. 1978. "A Tandemly Repeated Sequence at the Termini of the Extrachromosomal Ribosomal RNA Genes in Tetrahymena". *Journal of Molecular Biology* 120 (1): 33–53. [https://doi.org/10.1016/0022-2836\(78\)90294-2](https://doi.org/10.1016/0022-2836(78)90294-2).
- Boateng, Kingsley A., Marina A. Bellani, Ivan V. Gregoret, Florencia Pratto, and R. Daniel Camerini-Otero. 2013. "Homologous Pairing Preceding SPO11-Mediated Double-Strand Breaks in Mice". *Developmental Cell* 24 (2): 196–205. <https://doi.org/10.1016/j.devcel.2012.12.002>.
- Brown, D. D., P. C. Wensink, and E. Jordan. 1972. "A Comparison of the Ribosomal DNA's of *Xenopus laevis* and *Xenopus mulleri*: The Evolution of Tandem Genes". *Journal of Molecular Biology* 63 (1): 57–73. [https://doi.org/10.1016/0022-2836\(72\)90521-9](https://doi.org/10.1016/0022-2836(72)90521-9).
- Brown, W. R. A., P. J. MacKinnon, A. Villasanté, N. Spurr, V. J. Buckle, and M. J. Dobson. 1990. "Structure and Polymorphism of Human Telomere-Associated DNA". *Cell* 63 (1): 119–32. [https://doi.org/10.1016/0092-8674\(90\)90293-N](https://doi.org/10.1016/0092-8674(90)90293-N).
- Burgoyne, Paul S. 1982. "Genetic Homology and Crossing over in the X and Y Chromosomes of Mammals". *Human Genetics* 61 (2): 85–90. <https://doi.org/10.1007/BF00274192>.
- Cabianca, Daphne S., Valentina Casa, Beatrice Bodega, Alexandros Xynos, Enrico Ginelli, Yujiro Tanaka, and Davide Gabellini. 2012. "A Long ncRNA Links Copy Number Variation to a Polycomb/trithorax Epigenetic Switch in FSHD Muscular Dystrophy". *Cell* 149 (4): 819–31. <https://doi.org/10.1016/j.cell.2012.03.035>.
- Caspersson, T., L. Zech, C. Johansson, and E. J. Modest. 1970. "Identification of Human Chromosomes by DNA-Binding Fluorescent Agents". *Chromosoma* 30 (2): 215–27. <https://doi.org/10.1007/BF00282002>.
- Cechova, Monika, and others. 2025. "Telomere-to-Telomere Assembly of a Multi-Generation Human Pedigree". (*In Press*).
- Cechova, Monika, Glennis A. Hartley, Savannah J. Hoyt, Glennis A. Logsdon, DongAhn Yoo, Mitchell R. Vollger, Andrea Guarracino, et al. 2025. "Origin and Evolution of Acrocentric Chromosomes in Human and Great Apes". *Biorxiv*. <https://doi.org/10.1101/2025.12.22.696095>.
- Chaisson, Mark J. P., Ashley D. Sanders, Xuefang Zhao, Ankit Malhotra, David Porubsky, Tobias Rausch, Eugene J. Gardner, et al. 2019. "Multi-Platform Discovery of Haplotype-Resolved Structural Variation in Human Genomes". *Nature Communications* 10:1784. <https://doi.org/10.1038/s41467-018-08148-z>.
- Charchar, F. J., M. Svartman, N. El-Mogharbel, M. Ventura, P. Kirby, M. R. Matarazzo, A. Ciccodicola, M. Rocchi, M. D'Esposito, and J. A. M. Graves. 2003. "Complex Events in the Evolution of the Human Pseudoautosomal Region 2 (PAR2)". *Genome Research* 13 (2): 281–86. <https://doi.org/10.1101/gr.390503>.
- Chen, Jian-Min, David N. Cooper, Nadia Chuzhanova, Claude Férec, and George P. Patrinos. 2007. "Gene Conversion: Mechanisms, Evolution and Human Disease". *Nature Reviews Genetics* 8 (10): 762–75. <https://doi.org/10.1038/nrg2193>.
- Cheng, Haoyu, Gregory T. Concepcion, Xiaowen Feng, Haowen Zhang, and Heng Li. 2021. "Haplotype-Resolved De Novo Assembly Using Phased Assembly Graphs with Hifiasm". *Nature Methods* 18:170–75. <https://doi.org/10.1038/s41592-020-01056-5>.

- Chikashige, Yuji, Chihiro Tsutsumi, Miho Yamane, Kasumi Okamasa, Tokuko Haraguchi, and Yasushi Hiraoka. 2006. "Meiotic Proteins Bqt1 and Bqt2 Tether Telomeres to Form the Bouquet Arrangement of Chromosomes". *Cell* 125 (1): 59–69. <https://doi.org/10.1016/j.cell.2006.01.048>.
- Chu, Hsueh-Ping, John E. Froberg, Barry Kesner, Hyun Jung Oh, Fei Ji, Ruslan Sadreyev, Stefan F. Pinter, and Jeannie T. Lee. 2017. "PAR-TERRA Directs Homologous Sex Chromosome Pairing". *Nature Structural & Molecular Biology* 24 (8): 620–31. <https://doi.org/10.1038/nsmb.3432>.
- Ciccodicola, A., M. D'Esposito, T. Esposito, F. Gianfrancesco, C. Migliaccio, M. G. Miano, M. R. Matarazzo, et al. 2000. "Differentially Regulated and Evolved Genes in the Fully Sequenced Xq/Yq Pseudoautosomal Region". *Human Molecular Genetics* 9 (3): 395–401. <https://doi.org/10.1093/hmg/9.3.395>.
- Collins, Ryan L., Harrison Brand, Konrad J. Karczewski, Xuefang Zhao, Jessica Alföldi, Laurent C. Francioli, Amit V. Khera, et al. 2020. "A Structural Variation Reference for Medical and Population Genetics". *Nature* 581 (7809): 444–51. <https://doi.org/10.1038/s41586-020-2287-8>.
- Computational Pan-Genomics Consortium. 2018. "Computational Pan-Genomics: Status, Promises and Challenges". *Briefings in Bioinformatics* 19 (1): 118–35. <https://doi.org/10.1093/bib/bbw089>.
- Cotter, David J., Sarah M. Brotman, and Melissa A. Wilson Sayres. 2016. "Genetic Diversity on the Human X Chromosome Does Not Support a Strict Pseudoautosomal Boundary". *Genetics* 203 (1): 485–92. <https://doi.org/10.1534/genetics.114.172692>.
- Der-Sarkissian, Hera, Gilles Vergnaud, Yves-Marie Borde, Gérard Thomas, and J. Arturo Londono-Vallejo. 2002. "Segmental Polymorphisms in the Proterminal Regions of a Subset of Human Chromosomes". *Genome Research* 12 (11): 1673–78. <https://doi.org/10.1101/gr.322802>.
- Deutekom, J. C. van, E. Bakker, R. J. Lemmers, M. J. van der Wielen, E. Bik, M. H. Hofker, G. W. Padberg, and R. R. Frants. 1996. "Evidence for Subtelomeric Exchange of 3.3 Kb Tandemly Repeated Units between Chromosomes 4q35 and 10q26: Implications for Genetic Counselling and Etiology of FSHD1". *Human Molecular Genetics* 5 (12): 1997–2003. <https://doi.org/10.1093/hmg/5.12.1997>.
- Dover, Gabriel. 1986. "Molecular Drive in Multigene Families: How Biological Novelty Arise, Spread and Are Assimilated". *Trends in Genetics* 2:159–65. [https://doi.org/10.1016/0168-9525\(86\)90211-8](https://doi.org/10.1016/0168-9525(86)90211-8).
- Dover, Gabriel A. 1982. "Molecular Drive: A Cohesive Mode of Species Evolution". *Nature* 299 (5879): 111–17. <https://doi.org/10.1038/299111a0>.
- Duret, Laurent, and Nicolas Galtier. 2009. "Biased Gene Conversion and the Evolution of Mammalian Genomic Landscapes". *Annual Review of Genomics and Human Genetics* 10:285–311. <https://doi.org/10.1146/annurev-genom-082908-150001>.
- Ebert, Peter, Peter A. Audano, Qihui Zhu, Bernardo Rodriguez-Martin, David Porubsky, Marc Jan Bonder, Arvis Sulovari, et al. 2021. "Haplotype-Resolved Diverse Human Genomes and Integrated Analysis of Structural Variation". *Science* 372 (6537): eabf7117. <https://doi.org/10.1126/science.abf7117>.
- Eichler, Evan E. 2001. "Recent Duplication, Domain Accretion and the Dynamic Mutation of the Human Genome". *Trends in Genetics* 17 (11): 661–69. [https://doi.org/10.1016/S0168-9525\(01\)02492-1](https://doi.org/10.1016/S0168-9525(01)02492-1).
- Eickbush, Thomas H., and Danna G. Eickbush. 2007. "Finely Orchestrated Movements: Evolution of the Ribosomal RNA Genes". *Genetics* 175 (2): 477–85. <https://doi.org/10.1534/genetics.107.071399>.

- Eizenga, Jordan M., Adam M. Novak, Jonas A. Sibbesen, Simon Heumos, Ali Ghaffaari, Glenn Hickey, Xian Chang, et al. 2020. "Pangenome Graphs". *Annual Review of Genomics and Human Genetics* 21:139–62. <https://doi.org/10.1146/annurev-genom-120219-080406>.
- Ellis, N. A., P. J. Goodfellow, B. Pym, M. Smith, M. Palmer, A. M. Frischau, and P. N. Goodfellow. 1989. "The Pseudoautosomal Boundary in Man Is Defined by an Alu Repeat Sequence Inserted on the Y Chromosome". *Nature* 337 (6202): 81–84. <https://doi.org/10.1038/337081a0>.
- Ellis, N., A. Taylor, B. O. Bengtsson, J. Kidd, J. Rogers, and P. N. Goodfellow. 1990. "Population Structure of the Human Pseudoautosomal Boundary". *Nature* 344:663–65. <https://doi.org/10.1038/344663a0>.
- Filatov, Dmitry A. 2004. "A Gradient of Silent Substitution Rate in the Human Pseudoautosomal Region". *Molecular Biology and Evolution* 21 (2): 410–17. <https://doi.org/10.1093/molbev/msh032>.
- Flaquer, Antonia, Christine Fischer, and Thomas F. Wienker. 2009. "A New Sex-Specific Genetic Map of the Human Pseudoautosomal Regions (PAR1 and PAR2)". *Human Heredity* 68 (3): 192–200. <https://doi.org/10.1159/000224639>.
- Flaquer, Antonia, Gudrun A. Rappold, Thomas F. Wienker, and Christine Fischer. 2008. "The Human Pseudoautosomal Regions: A Review for Genetic Epidemiologists". *European Journal of Human Genetics* 16:771–79. <https://doi.org/10.1038/ejhg.2008.63>.
- Flint, J., G. P. Bates, K. Clark, A. Dorman, D. Willingham, B. A. Roe, G. Micklem, D. R. Higgs, and E. J. Louis. 1997. "Sequence Comparison of Human and Yeast Telomeres Identifies Structurally Distinct Subtelomeric Domains". *Human Molecular Genetics* 6 (8): 1305–13. <https://doi.org/10.1093/hmg/6.8.1305>.
- Floutsakou, Ioanna, Saumya Agrawal, Thong T. Nguyen, Cathal Seoighe, Austen R. D. Ganley, and Brian McStay. 2013. "The Shared Genomic Architecture of Human Nucleolar Organizer Regions". *Genome Research* 23 (12): 2003–12. <https://doi.org/10.1101/gr.157941.113>.
- Freije, D., C. Helms, M. S. Watson, and H. Donis-Keller. 1992. "Identification of a Second Pseudoautosomal Region near the Xq and Yq Telomeres". *Science* 258 (5089): 1784–87. <https://doi.org/10.1126/science.1465614>.
- Fukami, Maki, Yasuko Fujisawa, Hiroyuki Ono, Tomoko Jinno, and Tsutomu Ogata. 2020. "Human Spermatogenesis Tolerates Massive Size Reduction of the Pseudoautosomal Region". *Genome Biology and Evolution* 12 (11): 1961–64. <https://doi.org/10.1093/gbe/evaa168>.
- Galtier, Nicolas, Gwenaél Piganeau, Dominique Mouchiroud, and Laurent Duret. 2001. "GC-Content Evolution in Mammalian Genomes: The Biased Gene Conversion Hypothesis". *Genetics* 159 (2): 907–11. <https://doi.org/10.1093/genetics/159.2.907>.
- Ganley, Austen R. D., and Takehiko Kobayashi. 2007. "Highly Efficient Concerted Evolution in the Ribosomal DNA Repeats: Total rDNA Repeat Variation Revealed by Whole-Genome Shotgun Sequence Data". *Genome Research* 17 (2): 184–91. <https://doi.org/10.1101/gr.5457707>.
- Garrison, Erik, Andrea Guarracino, Simon Heumos, Flavia Villani, Zhigui Bao, Lorenzo Tattini, Joerg Hagmann, et al. 2024. "Building Pangenome Graphs". *Nature Methods* 21:2008–12. <https://doi.org/10.1038/s41592-024-02430-3>.
- Garrison, Erik, Jouni Sirén, Adam M. Novak, Glenn Hickey, Jordan M. Eizenga, Eric T. Dawson, William Jones, et al. 2018. "Variation Graph Toolkit Improves Read Mapping by Representing Genetic Variation in the Reference". *Nature Biotechnology* 36 (9): 875–79. <https://doi.org/10.1038/nbt.4227>.

- Gershman, Ariel, Michael E. G. Sauria, Xavi Guitart, Mitchell R. Vollger, Paul W. Hook, Savannah J. Hoyt, Miten Jain, et al. 2022. "Epigenetic Patterns in a Complete Human Genome". *Science* 376 (6588): eabj5089. <https://doi.org/10.1126/science.abj5089>.
- Glusman, Gustavo, Itai Yanai, Irit Rubin, and Doron Lancet. 2001. "The Complete Human Olfactory Subgenome". *Genome Research* 11 (5): 685–702. <https://doi.org/10.1101/gr.171001>.
- Goodfellow, P. J., S. M. Darling, N. S. Thomas, and P. N. Goodfellow. 1986. "A Pseudoautosomal Gene in Man". *Science* 234 (4777): 740–43. <https://doi.org/10.1126/science.2877492>.
- Graves, J. A. M., M. J. Wakefield, and R. Toder. 1998. "The Origin and Evolution of the Pseudoautosomal Regions of Human Sex Chromosomes". *Human Molecular Genetics* 7 (13): 1991–96. <https://doi.org/10.1093/hmg/7.13.1991>.
- Grigorev, Kirill, Jonathan Foox, Daniela Bezdan, Daniel Butler, Jared J. Luxton, Jake Reed, Miles J. McKenna, et al. 2021. "Haplotype Diversity and Sequence Heterogeneity of Human Telomeres". *Genome Research* 31 (7): 1269–79. <https://doi.org/10.1101/gr.274639.120>.
- Guarracino, Andrea, Silvia Buonaiuto, Leonardo G. de Lima, Tamara Potapova, Arang Rhie, Sergey Koren, Boris Rubinstein, et al. 2023. "Recombination between Heterologous Human Acrocentric Chromosomes". *Nature* 617:335–43. <https://doi.org/10.1038/s41586-023-05976-y>.
- Hallast, Pille, Peter Ebert, Mark Loftus, Feyza Yilmaz, Peter A. Audano, Glennis A. Logsdon, Marc Jan Bonder, Weichen Zhou, Wolfram Hopps, and others. 2023. "Assembly of 43 Human Y Chromosomes Reveals Extensive Complexity and Variation". *Nature* 621:355–64. <https://doi.org/10.1038/s41586-023-06425-6>.
- Halldórsson, Bjarni V., Gunnar Palsson, Olafur A. Stefansson, Hakon Jonsson, Marteinn T. Hardarson, Hannes P. Eggertsson, Bjarni Gunnarsson, et al. 2019. "Characterizing Mutagenic Effects of Recombination Through a Sequence-Level Genetic Map". *Science* 363 (6425): eaau1043. <https://doi.org/10.1126/science.aau1043>.
- Hartley, Glennis A., David Porubsky, Glennis A. Logsdon, Evan E. Eichler, Karen H. Miga, Andrea Guarracino, and Erik Garrison. 2026. "Biobank-Scale Genotyping of Robertsonian Translocations Reveals Hidden Structural Variation on the Human Acrocentric Chromosomes". *Biorxiv*. <https://doi.org/10.1101/2026.03.08.710242>.
- Hasin-Brumshtein, Yehudit, Doron Lancet, and Tsviya Olender. 2009. "Human Olfaction: From Genomic Variation to Phenotypic Diversity". *Trends in Genetics* 25 (4): 178–84. <https://doi.org/10.1016/j.tig.2009.02.002>.
- Hastings, P. J., James R. Lupski, Susan M. Rosenberg, and Grzegorz Ira. 2009. "Mechanisms of Change in Gene Copy Number". *Nature Reviews Genetics* 10 (8): 551–64. <https://doi.org/10.1038/nrg2593>.
- Henderson, A. S., D. Warburton, and K. C. Atwood. 1972. "Location of Ribosomal DNA in the Human Chromosome Complement". *Proceedings of the National Academy of Sciences USA* 69 (11): 3394–98. <https://doi.org/10.1073/pnas.69.11.3394>.
- Heumos, Simon, Andrea Guarracino, Jan-Niklas Schmelzle, Jiajie Li, Zhiru Zhang, Joerg Hagmann, Sven Nahnsen, Pjotr Prins, and Erik Garrison. 2024. "Pangenome Graph Layout by Path-Guided Stochastic Gradient Descent". *Bioinformatics* 40 (6): btae363. <https://doi.org/10.1093/bioinformatics/btae363>.

- Hinch, Anjali G., Nicolas Altemose, Naila Noor, Peter Donnelly, and Simon R. Myers. 2014. "Recombination in the Human Pseudoautosomal Region PAR1". *Plos Genetics* 10 (7): e1004503. <https://doi.org/10.1371/journal.pgen.1004503>.
- Hiraoka, Yasushi, and Abby F. Dernburg. 2009. "The SUN Rises on Meiotic Chromosome Dynamics". *Developmental Cell* 17 (5): 598–605. <https://doi.org/10.1016/j.devcel.2009.10.014>.
- Hsu, T. C. 1979. "Human and Mammalian Cytogenetics: An Historical Perspective". *Springer-Verlag*.
- Hurles, Matthew. 2004. "Gene Conversion Homogenizes the CMT1A Paralogous Repeats". *BMC Genomics* 2:11. <https://doi.org/10.1186/1471-2164-2-11>.
- Jarmuz-Szymczak, Malgorzata, Joanna Janiszewska, Krzysztof Szyfter, and Lisa G. Shaffer. 2014. "Narrowing the Localization of the Region Breakpoint in Most Frequent Robertsonian Translocations". *Chromosome Research* 22 (4): 517–32. <https://doi.org/10.1007/s10577-014-9439-3>.
- Jeffreys, Alec J., and Celia A. May. 2004. "Intense and Highly Localized Gene Conversion Activity in Human Meiotic Crossover Hot Spots". *Nature Genetics* 36 (2): 151–56. <https://doi.org/10.1038/ng1287>.
- Kong, Augustine, Gudmar Thorleifsson, Daniel F. Gudbjartsson, Gisli Masson, Asgeir Sigurdsson, Aslaug Jonasdottir, G. Bragi Walters, et al. 2010. "Fine-Scale Recombination Rate Differences between Sexes, Populations and Individuals". *Nature* 467 (7319): 1099–1103. <https://doi.org/10.1038/nature09525>.
- Kowaljew, Valeria, Alejandra Marcowycz, Eugénie Anseau, Céline B. Conde, Sandrine Sauvage, Céline Matteotti, Carolina Arias, et al. 2007. "The DUX4 Gene at the FSHD1A Locus Encodes a Pro-Apoptotic Protein". *Neuromuscular Disorders* 17 (8): 611–23. <https://doi.org/10.1016/j.nmd.2007.04.002>.
- Krystal, M., P. D'Eustachio, F. H. Ruddle, and N. Arnheim. 1981. "Human Nucleolus Organizers on Nonhomologous Chromosomes Can Share the Same Ribosomal Gene Variants". *Proceedings of the National Academy of Sciences USA* 78 (9): 5744–48. <https://doi.org/10.1073/pnas.78.9.5744>.
- Lahn, B. T., and D. C. Page. 1999. "Four Evolutionary Strata on the Human X Chromosome". *Science* 286 (5441): 964–67. <https://doi.org/10.1126/science.286.5441.964>.
- Lee, Chih-Ying, Henning F. Horn, Colin L. Stewart, Brian Burke, Needhi Bhalla, Abby F. Dernburg, Michael N. Conrad, and Michael E. Dresser. 2015. "Mechanism and Regulation of Rapid Telomere Prophase Movements in Mouse Meiotic Chromosomes". *Cell Reports* 11 (4): 551–63. <https://doi.org/10.1016/j.celrep.2015.03.045>.
- Lemmers, Richard J. L. F., Peggy de Kievit, Lodewijk Sandkuijl, George W. Padberg, Gert-Jan B. van Ommen, Rune R. Frants, and Silvére M. van der Maarel. 2002. "Facioscapulohumeral Muscular Dystrophy Is Uniquely Associated with One of the Two Variants of the 4q Subtelomere". *Nature Genetics* 32 (2): 235–36. <https://doi.org/10.1038/ng999>.
- Lemmers, Richard J. L. F., Patrick J. van der Vliet, Rinse Klooster, Sabrina Sacconi, Pilar Camano, Johannes G. Dauwerse, Lauren Snider, et al. 2010. "A Unifying Genetic Model for Facioscapulohumeral Muscular Dystrophy". *Science* 329 (5999): 1650–53. <https://doi.org/10.1126/science.1189044>.
- Li, Heng, Xiaowen Feng, and Chong Chu. 2020. "The Design and Construction of Reference Pangenome Graphs with Minigraph". *Genome Biology* 21:265. <https://doi.org/10.1186/s13059-020-02168-z>.

- Liao, Daiqing. 1999. "Concerted Evolution: Molecular Mechanism and Biological Implications". *American Journal of Human Genetics* 64 (1): 24–30. <https://doi.org/10.1086/302221>.
- Liao, Wen-Wei, Mobin Asri, Jana Ebler, Daniel Doerr, Marina Haukness, Glenn Hickey, Shuangjia Lu, et al. 2023. "A Draft Human Pangenome Reference". *Nature* 617:312–24. <https://doi.org/10.1038/s41586-023-05896-x>.
- Lichter, Peter, Choong-Jin C. Tang, Karl Call, Glenn Hermanson, Glen A. Evans, David Housman, and David C. Ward. 1990. "High-Resolution Mapping of Human Chromosome 11 by in Situ Hybridization with Cosmid Clones". *Science* 247 (4938): 64–69. <https://doi.org/10.1126/science.2294592>.
- Lieberman-Aiden, Erez, Nynke L. van Berkum, Louise Williams, Maxim Imakaev, Tobias Ragoczy, Agnes Telling, Ido Amit, et al. 2009. "Comprehensive Mapping of Long-Range Interactions Reveals Folding Principles of the Human Genome". *Science* 326 (5950): 289–93. <https://doi.org/10.1126/science.1181369>.
- Lima, Leonardo G. de, Andrea Guarracino, and others. 2025. "Structural Variation and Breakpoint Analysis of the Human Robertsonian Translocations". (*In Submission*).
- Linardopoulou, Elena V., Eleanor M. Williams, Yuxin Fan, Cynthia Williams, Jeffrey A. Bailey, and Barbara J. Trask. 2005. "Human Subtelomeres Are Hot Spots of Interchromosomal Recombination and Segmental Duplication". *Nature* 437 (7055): 94–100. <https://doi.org/10.1038/nature04029>.
- Liu, Yichen Henry, Can Luo, Staunton G. Golding, Jacob B. Ioffe, and Xin Maizie Zhou. 2024. "Tradeoffs in Alignment and Assembly-Based Methods for Structural Variant Detection with Long-Read Sequencing Data". *Nature Communications* 15:2447. <https://doi.org/10.1038/s41467-024-46614-z>.
- Llorente, Bertrand, Catherine E. Smith, and Lorraine S. Symington. 2008. "Break-Induced Replication: What Is It and What Is It For?". *Cell Cycle* 7 (7): 859–64. <https://doi.org/10.4161/cc.7.7.5613>.
- Logsdon, Glennis A., Mitchell R. Vollger, PingHsun Hsieh, Yupeng Mao, Mikhail A. Liskovych, Sergey Koren, Sergey Nurk, et al. 2021. "The Structure, Function and Evolution of a Complete Human Chromosome 8". *Nature* 593:101–7. <https://doi.org/10.1038/s41586-021-03420-7>.
- Lovett, Susan T. 2004. "Encoded Errors: Mutations and Rearrangements Mediated by Misalignment at Repetitive DNA Sequences". *Molecular Microbiology* 52 (5): 1243–53. <https://doi.org/10.1111/j.1365-2958.2004.04076.x>.
- Lupski, James R. 1998. "Genomic Disorders: Structural Features of the Genome Can Lead to DNA Rearrangements and Human Disease Traits". *Trends in Genetics* 14 (10): 417–22. [https://doi.org/10.1016/S0168-9525\(98\)01555-8](https://doi.org/10.1016/S0168-9525(98)01555-8).
- Lupski, James R. 2009. "Genomic Disorders Ten Years On". *Genome Medicine* 1 (4): 42. <https://doi.org/10.1186/gm42>.
- Lupski, James R., and Pawel Stankiewicz. 2005. "Genomic Disorders: Molecular Mechanisms for Rearrangements and Conveyed Phenotypes". *Plos Genetics* 1 (6): e49. <https://doi.org/10.1371/journal.pgen.0010049>.
- Mallick, Swapam, Heng Li, Mark Lipson, Iain Mathieson, Melissa Gymrek, Fernando Racimo, Mengyao Zhao, et al. 2016. "The Simons Genome Diversity Project: 300 Genomes from 142 Diverse Populations". *Nature* 538 (7624): 201–6. <https://doi.org/10.1038/nature18964>.

- Mangs, A. Helena, and Brian J. Morris. 2007. "The Human Pseudoautosomal Region (PAR): Origin, Function and Future". *Current Genomics* 8 (2): 129–36. <https://doi.org/10.2174/138920207780368141>.
- Masny, Pawel S., Ulla Bengtsson, Suk-Au Chung, Jodi H. Martin, Baziél van Engelen, Silvère M. van der Maarel, and Sara T. Winokur. 2004. "Localization of 4q35.2 to the Nuclear Periphery: Is FSHD a Nuclear Envelope Disease?". *Human Molecular Genetics* 13 (17): 1857–71. <https://doi.org/10.1093/hmg/ddh205>.
- McStay, Brian. 2016. "Nucleolar Organizer Regions: Genomic ``dark Matter" Requiring Illumination". *Genes & Development* 30 (14): 1598–1610. <https://doi.org/10.1101/gad.283838.116>.
- Mefford, Heather C., and Barbara J. Trask. 2002. "The Complex Structure and Dynamic Evolution of Human Subtelomeres". *Nature Reviews Genetics* 3 (2): 91–102. <https://doi.org/10.1038/nrg727>.
- Mensah, Martin A., Matthew S. Hestand, Maarten H. D. Larmuseau, M. Isrie, N. Vanderheyden, M. Declercq, H. Van Esch, K. Devriendt, J. R. Vermeesch, and G. Froyen. 2014. "Pseudoautosomal Region 1 Length Polymorphism in the Human Population". *Plos Genetics* 10 (11): e1004578. <https://doi.org/10.1371/journal.pgen.1004578>.
- Miga, Karen H., Sergey Koren, Arang Rhie, Mitchell R. Vollger, Ariel Gershman, Andrey Bzikadze, Shelise Brooks, et al. 2020. "Telomere-to-Telomere Assembly of a Complete Human X Chromosome". *Nature* 585:79–84. <https://doi.org/10.1038/s41586-020-2547-7>.
- Monteiro, Bruno, Miguel Arenas, Maria Joao Prata, and Antonio Amorim. 2021. "Evolutionary Dynamics of the Human Pseudoautosomal Regions". *Plos Genetics* 17 (4): e1009532. <https://doi.org/10.1371/journal.pgen.1009532>.
- Nagano, Takashi, Yaniv Lubling, Tim J. Stevens, Stefan Schoenfelder, Eitan Yaffe, Wendy Dean, Ernest D. Laue, Amos Tanay, and Peter Fraser. 2013. "Single-Cell Hi-C Reveals Cell-to-Cell Variability in Chromosome Structure". *Nature* 502 (7469): 59–64. <https://doi.org/10.1038/nature12593>.
- Nei, Masatoshi, and Alejandro P. Rooney. 2005. "Concerted and Birth-and-Death Evolution of Multigene Families". *Annual Review of Genetics* 39:121–52. <https://doi.org/10.1146/annurev.genet.39.073003.112240>.
- Niimura, Yoshihito, and Masatoshi Nei. 2003. "Evolution of Olfactory Receptor Genes in the Human Genome". *Proceedings of the National Academy of Sciences* 100 (21): 12235–40. <https://doi.org/10.1073/pnas.1635157100>.
- Niimura, Yoshihito, Atsushi Matsui, and Kazushige Touhara. 2014. "Extreme Expansion of the Olfactory Receptor Gene Repertoire in African Elephants and Evolutionary Dynamics of Orthologous Gene Groups in 13 Placental Mammals". *Genome Research* 24 (9): 1485–96. <https://doi.org/10.1101/gr.169532.113>.
- Nurk, Sergey, Sergey Koren, Arang Rhie, Mikko Rautiainen, Andrey V. Bzikadze, Alla Mikheenko, Mitchell R. Vollger, et al. 2022. "The Complete Sequence of a Human Genome". *Science* 376 (6588): 44–53. <https://doi.org/10.1126/science.abj6987>.
- Olender, Tsviya, Doron Lancet, and Daniel W. Nebert. 2008. "Update on the Olfactory Receptor (OR) Gene Superfamily". *Human Genomics* 3 (1): 87–97. <https://doi.org/10.1186/1479-7364-3-1-87>.
- Ottaviani, Alexandre, Sylvie Rival-Gervier, Amina Boussouar, Andrea M. Foerster, Délia Rondier, Sabrina Sacconi, Claude Desnuelle, Eric Gilson, and Frederique Magdinier. 2009b. "The D4Z4

- Macrosatellite Repeat Acts as a CTCF and A-Type Lamins-Dependent Insulator in Facio-Scapulo-Humeral Dystrophy". *Plos Genetics* 5 (2): e1000394. <https://doi.org/10.1371/journal.pgen.1000394>.
- Ottaviani, Alexandre, Caroline Schluth-Bolard, Sylvie Rival-Gervier, Amina Boussouar, Délia Rondier, Andrea M. Foerster, Jérôme Morere, et al. 2009a. "Identification of a Perinuclear Positioning Element in Human Subtelomeres That Requires A-Type Lamins and CTCF". *EMBO Journal* 28 (16): 2428–36. <https://doi.org/10.1038/emboj.2009.201>.
- Page, S. L., and L. G. Shaffer. 1996. "Identification of a Common Breakpoint at 14q21q from De Novo and Familial Robertsonian Translocations: Breakpoint Diversity Illustrates Distinct Mechanisms for Robertsonian Translocation Formation". *Human Molecular Genetics* 5 (9): 1279–88. <https://doi.org/10.1093/hmg/5.9.1279>.
- Patel, Linda, Roger Kang, Scott C. Rosenberg, Yu Qiu, Ramya Raviram, Sora Chee, Rong Hu, Bing Ren, Francesca Cole, and Kevin D. Corbett. 2019. "Dynamic Reorganization of the Genome Shapes the Recombination Landscape in Meiotic Prophase". *Nature Structural & Molecular Biology* 26 (3): 164–74. <https://doi.org/10.1038/s41594-019-0187-0>.
- Penkner, Alexandra M., Alexander Fridkin, Jiradet Gloggnitzer, Antoine Baudrimont, Thomas Machacek, Alexander Woglar, Edina Csaszar, et al. 2009. "Meiotic Chromosome Homology Search Involves Modifications of the Nuclear Envelope Protein Matefin/SUN-1". *Cell* 139 (5): 920–33. <https://doi.org/10.1016/j.cell.2009.10.045>.
- Pinkel, Daniel, Tore Straume, and Joe W. Gray. 1986. "Cytogenetic Analysis Using Quantitative, High-Sensitivity, Fluorescence Hybridization". *Proceedings of the National Academy of Sciences USA* 83 (9): 2934–38. <https://doi.org/10.1073/pnas.83.9.2934>.
- Poriswanish, Nitikorn, James Eales, Xiaoguang Xu, David Scannali, Rita Neumann, Jon H. Wetton, Maciej Tomaszewski, Mark A. Jobling, and Celia A. May. 2025. "Multiple Origins and Phenotypic Implications of an Extended Human Pseudoautosomal Region Shown by Analysis of the UK Biobank". *American Journal of Human Genetics* 112 (4): 927–39. <https://doi.org/10.1016/j.ajhg.2025.01.026>.
- Poriswanish, Nitikorn, Rita Neumann, Jon H. Wetton, James Wagstaff, Maarten H. D. Larmuseau, Mark A. Jobling, and Celia A. May. 2018. "Recombination Hotspots in an Extended Human Pseudoautosomal Domain Predicted from Double-Strand Break Maps and Characterized by Sperm-Based Crossover Analysis". *Plos Genetics* 14 (10): e1007680. <https://doi.org/10.1371/journal.pgen.1007680>.
- Porubsky, David, and others. 2025. "Phased Genome Assemblies of the CEPH 1463 Four-Generation Pedigree Enable High-Resolution Analysis of Human Genome Variation". (*In Press*).
- Pritchard, C. A., P. J. Goodfellow, and P. N. Goodfellow. 1987. "Mapping the Limits of the Human Pseudoautosomal Region and a Candidate Sequence for the Male-Determining Gene". *Nature* 328:273–75. <https://doi.org/10.1038/328273a0>.
- Ramani, Vijay, Xinxian Deng, Ruolan Qiu, Kevin L. Gunderson, Frank J. Steemers, Christine M. Disteche, William S. Noble, Zhijun Duan, and Jay Shendure. 2017. "Massively Multiplex Single-Cell Hi-C". *Nature Methods* 14 (3): 263–66. <https://doi.org/10.1038/nmeth.4155>.
- Rao, Suhas S. P., Miriam H. Huntley, Neva C. Durand, Elena K. Stamenova, Ivan D. Bochkov, James T. Robinson, Adrian L. Sanborn, et al. 2014. "A 3D Map of the Human Genome at Kilobase Resolution Reveals Principles of Chromatin Looping". *Cell* 159 (7): 1665–80. <https://doi.org/10.1016/j.cell.2014.11.021>.

- Rautiainen, Mikko, Sergey Nurk, Brian P. Walenz, Glennis A. Logsdon, David Porubsky, Arang Rhie, Evan E. Eichler, Adam M. Phillippy, and Sergey Koren. 2023. "Telomere-to-Telomere Assembly of Diploid Chromosomes with Verkko". *Nature Biotechnology* 41:1474–82. <https://doi.org/10.1038/s41587-023-01662-6>.
- Rhie, Arang, Sergey Nurk, Sergey Koren, Mikko Rautiainen, Monika Cechova, Savannah J. Hoyt, Dylan J. Taylor, Nicolas Altemose, Paul W. Hook, and others. 2023. "The Complete Sequence of a Human Y Chromosome". *Nature* 621:344–54. <https://doi.org/10.1038/s41586-023-06457-y>.
- Riethman, Harold. 2008. "Human Subtelomeric Copy Number Variations". *Cytogenetic and Genome Research* 123 (1–4): 244–52. <https://doi.org/10.1159/000184714>.
- Riethman, Harold C., Zhihai Xiang, Sheila Paul, Eve Morse, Xiao-Lin Hu, Jonathan Flint, Hai-Chien Chi, Donald L. Grady, and Robert K. Moyzis. 2001. "Integration of Telomere Sequences with the Draft Human Genome Sequence". *Nature* 409 (6822): 948–51. <https://doi.org/10.1038/35057180>.
- Riethman, Harold, Anthony Ambrosini, Carlos Castaneda, Joshua Finklestein, Xiao-Lin Hu, Uma Mudunuri, Sheila Paul, and Jeff Wei. 2004. "Mapping and Initial Analysis of Human Subtelomeric Sequence Assemblies". *Genome Research* 14 (1): 18–28. <https://doi.org/10.1101/gr.1245004>.
- Robertson, William Rees Brebner. 1916. "Chromosome Studies. I. Taxonomic Relationships Shown in the Chromosomes of Tettigidae and Acrididae: V-Shaped Chromosomes and Their Significance in Acrididae, Locustidae, And Gryllidae: Chromosomes and Variation". *Journal of Morphology* 27 (2): 179–331. <https://doi.org/10.1002/jmor.1050270202>.
- Ross, Mark T., Darren V. Grafham, Alison J. Coffey, and others. 2005. "The DNA Sequence of the Human X Chromosome". *Nature* 434 (7031): 325–37. <https://doi.org/10.1038/nature03440>.
- Rouyer, F., M. C. Simmler, C. Johnsson, G. Vergnaud, H. J. Cooke, and J. Weissenbach. 1986. "A Gradient of Sex Linkage in the Pseudoautosomal Region of the Human Sex Chromosomes". *Nature* 319 (6051): 291–95. <https://doi.org/10.1038/319291a0>.
- Rudd, M. Katharine, Cynthia Friedman, Sean S. Parghi, Elena V. Linardopoulou, Li Hsu, and Barbara J. Trask. 2007. "Elevated Rates of Sister Chromatid Exchange at Chromosome Ends". *Plos Genetics* 3 (2): e32. <https://doi.org/10.1371/journal.pgen.0030032>.
- Ruiz-Herrera, Aurora, Solomon G. Nergadze, Marco Santagostino, and Elena Giulotto. 2008. "Telomeric Repeats Far from the Ends: Mechanisms of Origin and Role in Evolution". *Cytogenetic and Genome Research* 122 (3–4): 219–28. <https://doi.org/10.1159/000167807>.
- Sasani, Thomas A., Brent S. Pedersen, Ziyue Gao, Lisa Baird, Molly Przeworski, Lynn B. Jorde, and Aaron R. Quinlan. 2019. "Large, Three-Generation Human Families Reveal Post-Zygotic Mosaicism and Variability in Germline Mutation Accumulation". *Elife* 8:e46922. <https://doi.org/10.7554/eLife.46922>.
- Scherthan, Harry, Susanne Weich, Helmut Schwegler, Christa Heyting, Michael Härle, and Thomas Cremer. 1996. "Centromere and Telomere Movements during Early Meiotic Prophase of Mouse and Man Are Associated with the Onset of Chromosome Pairing". *Journal of Cell Biology* 134 (5): 1109–25. <https://doi.org/10.1083/jcb.134.5.1109>.
- Schmickel, R. D. 1973. "Quantitation of Human Ribosomal DNA: Hybridization of Human DNA with Ribosomal RNA for Quantitation and Fractionation". *Pediatric Research* 7 (1): 5–12. <https://doi.org/10.1203/00006450-197301000-00002>.

- Shao, Haojing, Chenxi Zhou, Minh Duc Cao, and Lachlan J. M. Coin. 2018. "Ongoing Human Chromosome End Extension Revealed by Analysis of BioNano and Nanopore Data". *Scientific Reports* 8:16616. <https://doi.org/10.1038/s41598-018-34774-0>.
- Sharp, Andrew J., Sierra Hansen, Rebecca R. Selzer, Ze Cheng, Regina Regan, Jane A. Hurst, Helen Stewart, et al. 2006. "Discovery of Previously Unidentified Genomic Disorders from the Duplication Architecture of the Human Genome". *Nature Genetics* 38 (9): 1038–42. <https://doi.org/10.1038/ng1862>.
- Skaletsky, Helen, Tomoko Kuroda-Kawaguchi, Patrick J. Minx, Holland S. Cordum, LaDeana Hillier, Laura G. Brown, Sjoerd Repping, et al. 2003. "The Male-Specific Region of the Human Y Chromosome Is a Mosaic of Discrete Sequence Classes". *Nature* 423 (6942): 825–37. <https://doi.org/10.1038/nature01722>.
- Sluis, Marjolein van, Máire Ó. Gailín, Joseph G. W. McCarter, Hazel Mangan, Alice Grob, and Brian McStay. 2019. "Human NORs, Comprising rDNA Arrays and Functionally Conserved Distal Elements, Are Located within Dynamic Chromosomal Regions". *Genes & Development* 33 (23–24): 1688–1701. <https://doi.org/10.1101/gad.331892.119>.
- Smith, G. P. 1976. "Evolution of Repeated DNA Sequences by Unequal Crossover". *Science* 191 (4227): 528–35. <https://doi.org/10.1126/science.1251186>.
- Snider, Lauren, Linda N. Geng, Richard J. L. F. Lemmers, Michael Kyba, Carol B. Ware, Andrew M. Nelson, Rabi Tawil, et al. 2010. "Facioscapulohumeral Dystrophy: Incomplete Suppression of a Retrotransposed Gene". *Plos Genetics* 6 (10): e1001181. <https://doi.org/10.1371/journal.pgen.1001181>.
- Sosa, Brandon A., Andrea Rothballer, Ulrike Kutay, and Thomas U. Schwartz. 2012. "LINC Complexes Form by Binding of Three KASH Peptides to Domain Interfaces of Trimeric SUN Proteins". *Cell* 149 (5): 1035–47. <https://doi.org/10.1016/j.cell.2012.03.046>.
- Stankiewicz, Pawel, and James R. Lupski. 2002. "Genome Architecture, Rearrangements and Genomic Disorders". *Trends in Genetics* 18 (2): 74–82. [https://doi.org/10.1016/S0168-9525\(02\)02592-1](https://doi.org/10.1016/S0168-9525(02)02592-1).
- Stankiewicz, Pawel, and James R. Lupski. 2010. "Structural Variation in the Human Genome and Its Role in Disease". *Annual Review of Medicine* 61:437–55. <https://doi.org/10.1146/annurev-med-100708-204735>.
- Stergachis, Andrew B., Brian M. Debo, Eric Haugen, L. Stirling Churchman, and John A. Stamatoyannopoulos. 2020. "Single-Molecule Regulatory Architectures Captured by Chromatin Fiber Sequencing". *Science* 368 (6498): 1449–54. <https://doi.org/10.1126/science.aaz1646>.
- Stevens, Tim J., David Lando, Srinjan Basu, Liam P. Atkinson, Yang Cao, Steven F. Lee, Martin Leeb, et al. 2017. "3D Structures of Individual Mammalian Genomes Studied by Single-Cell Hi-C". *Nature* 544 (7648): 59–64. <https://doi.org/10.1038/nature21429>.
- Stimpson, Kaitlin M., Lori L. Sullivan, Molly E. Kuo, and Beth A. Sullivan. 2014. "Nucleolar Organization, Ribosomal DNA Array Stability, And Acrocentric Chromosome Integrity Are Linked to Telomere Function". *PLOS ONE* 9 (3): e92432. <https://doi.org/10.1371/journal.pone.0092432>.
- Stong, Nicholas, Zhong Deng, Ravi Gupta, Sufeng Hu, Sheila Paul, Aaron K. Weiner, Evan E. Eichler, et al. 2014. "Subtelomeric CTCF and Cohesin Binding Site Organization Using Improved Subtelomere Assemblies and a Novel Annotation Pipeline". *Genome Research* 24 (6): 1039–50. <https://doi.org/10.1101/gr.166983.113>.

- Stout, K., S. van der Maarel, R. R. Frants, G. W. Padberg, H. H. Ropers, and T. Haaf. 1999. "Somatic Pairing between Subtelomeric Chromosome Regions: Implications for Human Genetic Disease?". *Chromosome Research* 7 (5): 323–29. <https://doi.org/10.1023/A:1009287518015>.
- Stults, Debra M., Mark W. Killen, Heather H. Pierce, and Andrew J. Pierce. 2008. "Genomic Architecture and Inheritance of Human Ribosomal RNA Gene Clusters". *Genome Research* 18 (1): 13–18. <https://doi.org/10.1101/gr.6858507>.
- Sudmant, Peter H., Tobias Rausch, Eugene J. Gardner, Robert E. Handsaker, Alexej Abyzov, John Huddleston, Yan Zhang, et al. 2015. "An Integrated Map of Structural Variation in 2,504 Human Genomes". *Nature* 526 (7571): 75–81. <https://doi.org/10.1038/nature15394>.
- Tan, Longzhi, Dong Xing, Chi-Han Chang, Heng Li, and X. Sunney Xie. 2018. "Three-Dimensional Genome Structures of Single Diploid Human Cells". *Science* 361 (6405): 924–28. <https://doi.org/10.1126/science.aat5641>.
- Therman, Eeva, Bonnie Susman, and Carter Denniston. 1989. "The Nonrandom Participation of Human Acrocentric Chromosomes in Robertsonian Translocations". *Annals of Human Genetics* 53 (1): 49–65. <https://doi.org/10.1111/j.1469-1809.1989.tb01121.x>.
- Trask, Barbara J. 1991. "Fluorescence in Situ Hybridization: Applications in Cytogenetics and Gene Mapping". *Trends in Genetics* 7 (5): 149–54. [https://doi.org/10.1016/0168-9525\(91\)90103-W](https://doi.org/10.1016/0168-9525(91)90103-W).
- Trask, Barbara J., Cynthia Friedman, Antonio Martin-Gallardo, Lee Rowen, Catherine Akinbami, Janelle Blankenship, Cassandra Collins, et al. 1998. "Members of the Olfactory Receptor Gene Family Are Contained in Large Blocks of DNA Duplicated Polymorphically near the Ends of Human Chromosomes". *Human Molecular Genetics* 7 (1): 13–26. <https://doi.org/10.1093/hmg/7.1.13>.
- Ulahannan, Netha, Matthew Pendleton, Aditya Deshpande, Stefan Schwenk, Jacob M. Behr, Xiaoguang Dai, Carly Tyler, et al. 2019. "Nanopore Sequencing of DNA Concatemers Reveals Higher-Order Features of Chromatin Structure". *Biorxiv*. <https://doi.org/10.1101/833590>.
- Vollger, Mitchell R., Xavi Guitart, Philip C. Dishuck, Ludovica Mercuri, William T. Harvey, Ariel Gershman, Mark Diekhans, et al. 2023. "Segmental Duplications and Their Variation in a Complete Human Genome". *Science* 376 (6588): eabj6965. <https://doi.org/10.1126/science.abj6965>.
- Wagner, Justin, Nathan D. Olson, Lindsay Harris, Ziad Khan, Jan Farek, Medhat Mahmoud, Adrian Stankovic, et al. 2022. "Benchmarking Challenging Small Variants with Linked and Long Reads". *Cell Genomics* 2 (5): 100128. <https://doi.org/10.1016/j.xgen.2022.100128>.
- Wijmenga, Cisca, Jane E. Hewitt, Lodewijk A. Sandkuijl, Linda N. Clark, Tracy J. Wright, Hans G. Dauwerse, Anne-Marie Gruter, et al. 1992. "Chromosome 4q DNA Rearrangements Associated with Facioscapulohumeral Muscular Dystrophy". *Nature Genetics* 2 (1): 26–30. <https://doi.org/10.1038/ng0992-26>.
- Wilkie, A. O. M., D. R. Higgs, K. A. Rack, V. J. Buckle, N. K. Spurr, N. Fischel-Ghodsian, I. Ceccherini, W. R. A. Brown, and P. C. Harris. 1991. "Stable Length Polymorphism of up to 260 Kb at the Tip of the Short Arm of Human Chromosome 16". *Cell* 64 (3): 595–606. [https://doi.org/10.1016/0092-8674\(91\)90243-R](https://doi.org/10.1016/0092-8674(91)90243-R).
- Williams, Amy L., Giulio Genovese, Thomas Dyer, Nicolas Altemose, Katherine Truax, Goo Jun, Nick Patterson, et al. 2015. "Non-Crossover Gene Conversions Show Strong GC Bias and Unexpected Clustering in Humans". *Elife* 4:e4637. <https://doi.org/10.7554/eLife.04637>.

- Worton, Ronald G., Janet Sutherland, John E. Sylvester, Huntington F. Willard, Serge Bodrug, Isabelle Dubé, Callum Duff, Victoria Kean, Peter N. Ray, and Richard D. Schmickel. 1988. "Human Ribosomal RNA Genes: Orientation of the Tandem Array and Conservation of the 5' End". *Science* 239 (4835): 64–68. <https://doi.org/10.1126/science.3336775>.
- Xu, Hanbo, and others. 2025. "Three-Dimensional Genome Organization of Human Sperm at Single-Cell Resolution". *Nature Communications* 16:3805. <https://doi.org/10.1038/s41467-025-58889-x>.
- Young, Eleanor, Heba Z. Abid, Pui-Yan Kwok, Harold Riethman, and Ming Xiao. 2020. "Comprehensive Analysis of Human Subtelomeres by Whole Genome Mapping". *Plos Genetics* 16 (1): e1008347. <https://doi.org/10.1371/journal.pgen.1008347>.
- Zickler, Denise, and Nancy Kleckner. 1998. "The Leptotene-Zygotene Transition of Meiosis". *Annual Review of Genetics* 32:619–97. <https://doi.org/10.1146/annurev.genet.32.1.619>.
- Zickler, Denise, and Nancy Kleckner. 2015. "Recombination, Pairing, And Synapsis of Homologs during Meiosis". *Cold Spring Harbor Perspectives in Biology* 7 (6): a16626. <https://doi.org/10.1101/cshperspect.a016626>.
- Zook, Justin M., Nancy F. Hansen, Nathan D. Olson, Lindsay Chapman, James C. Mullikin, Chunlin Xiao, Stephen Sherry, et al. 2020. "A Robust Benchmark for Germline Structural Variant Detection". *Biorxiv*. <https://doi.org/10.1101/664623>.
- Zuo, Wu, Guangyu Chen, Zhongjun Gao, Shuai Li, Yanyan Chen, Chenhua Huang, Juan Chen, Zhenhai Chen, Ming Lei, and Qian Bian. 2021. "Stage-Resolved Hi-C Analyses Reveal Meiotic Chromosome Organizational Features Influencing Homolog Alignment". *Nature Communications* 12 (1): 5827. <https://doi.org/10.1038/s41467-021-26033-0>.